ECON 803 Microeconomic Theory I

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Question-only Practice Book - Fall 2026

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How to Use This Practice Book

This volume collects the complete student-facing problem program for ECON 803. It contains 91 chapter problems and 10 additional practice problems.

Problem identifiers are permanent catalog labels. They can differ from the printed chapter number after topics are combined, and they remain unchanged in weekly assignments, case studies and the instructor handbook.

Part I problems are compact benchmarks intended for derivation and discussion by hand. Part II problems develop proofs, applications, and one named use case for each bank topic. The three general equilibrium topics share one printed chapter. Core, Proof and Applied identify the kind of work. The Advanced sections add integrated exam practice on production and equilibrium.

Before using Codex, Python, or Lean, record four lines: Seed identifies the hand result being extended; Fixed records the assumptions that remain unchanged; Scaled names the single enlarged dimension or strengthened claim; Risk states the new failure mode. The resulting artifact must reproduce the hand case and include a boundary or deliberately failing test.

A successful computation or compilation verifies only the statement encoded in the artifact. Students remain responsible for the economic assumptions, the interpretation, and the distinction between numerical evidence and proof.

Fall 2026 Weekly Practice Map

This map organizes the non-case chapter problems into thirteen weekly sets. Chapter use cases remain a separate case-study stream. Reserve problems are optional additional practice. The stable identifiers below refer to the full questions later in this book.

PS01: Utility maximization and demand

Class meeting: Sep. 11, 2026. Due: Sep. 18, 2026.

Required: 1.1, 1.2, 1.4. Formal or applied extension: 1.3, 1.6. Optional reserve: None.

PS02: Duality and welfare

Class meeting: Sep. 18, 2026. Due: Sep. 25, 2026.

Required: 2.1, 2.3, 2.5. Formal or applied extension: 2.2, 2.4. Optional reserve: None.

PS03: Revealed preference and recoverability

Class meeting: Sep. 25, 2026. Due: Oct. 2, 2026.

Required: 3.1, 3.3, 3.6. Formal or applied extension: 3.2, 3.5. Optional reserve: R001.

PS04: Abstract choice and order foundations

Class meeting: Oct. 2, 2026. Due: Oct. 9, 2026.

Required: 4.1, 4.3, A.1. Formal or applied extension: A.2. Optional reserve: R002, R004.

PS05: Expected utility and risk

Class meeting: Oct. 9, 2026. Due: Oct. 16, 2026.

Required: 5.1, 5.2. Formal or applied extension: 5.3. Optional reserve: None.

PS06: Stochastic choice and attention

Class meeting: Oct. 16, 2026. Due: Oct. 23, 2026.

Required: 6.1, 6.3, 7.1. Formal or applied extension: 7.2. Optional reserve: R003.

PS07: Aggregation

Class meeting: Oct. 23, 2026. Due: Oct. 30, 2026.

Required: 8.1, 8.2. Formal or applied extension: None. Optional reserve: None.

PS08: Production and cost

Class meeting: Oct. 30, 2026. Due: Nov. 6, 2026.

Required: 9.1, 9.3. Formal or applied extension: 9.2. Optional reserve: R005.

Advanced exam preparation: 9.A5, 9.A2. Further practice: 9.A1, 9.A3, 9.A4, 9.A6. Attempt the recommended pair by hand after the foundations; keep the rest for revision. This is additional preparation unless assigned in class.

PS09: Competitive markets and exchange

Class meeting: Nov. 6, 2026. Due: Nov. 13, 2026.

Required: 10.1, 10.2, 11.1. Formal or applied extension: 10.3. Optional reserve: R006.

Advanced exam preparation: 10.A5, 10.A6. Further practice: 10.A1, 10.A2, 10.A3, 10.A4, 10.A7, 10.A8. Attempt the recommended pair by hand after the foundations; keep the rest for revision. This is additional preparation unless assigned in class.

PS10: Welfare and positive equilibrium

Class meeting: Nov. 13, 2026. Due: Nov. 20, 2026.

Required: 11.2, 12.1, 12.2. Formal or applied extension: 11.3, 12.3. Optional reserve: R008, R009.

Advanced exam preparation: 11.A1, 11.A7. Further practice: 11.A2, 11.A3, 11.A4. Attempt the recommended pair by hand after the foundations; keep the rest for revision. This is additional preparation unless assigned in class.

PS11: Empirical equilibrium and asset markets

Class meeting: Nov. 20, 2026. Due: Nov. 27, 2026.

Required: 13.1, 14.1, 14.2. Formal or applied extension: 13.2. Optional reserve: R007, R010.

Advanced exam preparation: 11.A5, 11.A6. Further practice: 12.A1. Attempt the recommended pair by hand after the foundations; keep the rest for revision. This is additional preparation unless assigned in class.

PS12: Matching

Class meeting: Nov. 27, 2026. Due: Dec. 4, 2026.

Required: 15.1, 15.2, A.3. Formal or applied extension: None. Optional reserve: None.

PS13: Computable equilibrium

Class meeting: Dec. 4, 2026. Due: Dec. 7, 2026.

Required: 16.1, 16.2. Formal or applied extension: None. Optional reserve: None.

Advanced exam preparation: 16.A2, 16.A3. Further practice: 16.A1. Attempt the recommended pair by hand after the foundations; keep the rest for revision. This is additional preparation unless assigned in class.

Part I

The Classical Rational Consumer

Chapter 1

Utility Maximization and Consumer Demand

Part I: By hand

Problem 1.1 [Core]. Budget scaling and Cobb-Douglas demand.

Let $X = \mathbb{R}^2_+$ and $B(p, w) = \{x \in X : p \cdot x \leq w\}$, where $p \gg 0$ and $w > 0$.

(a) Prove that $B(\lambda p, \lambda w) = B(p, w)$ for every $\lambda > 0$. Explain why every demand correspondence generated by preference maximization is homogeneous of degree zero in (p, w).

(b) Suppose preferences are locally nonsatiated. Prove that every chosen bundle satisfies Walras' law, $p \cdot x = w$.

(c) For $u(x) = x_1^{\alpha} x_2^{1-\alpha}$ with $0 < \alpha < 1$, derive Marshallian demand and verify both properties directly.

Problem 1.2 [Core]. CES curvature, demand, and expenditure shares.

For $\rho < 1$ and $\rho \neq 0$, let $u_\rho(x) = (\frac{1}{2}x_1^\rho + \frac{1}{2}x_2^\rho)^{1/\rho}$, with the Cobb-Douglas limit used at $\rho = 0$ and the boundary extension stated in the text. Write $\sigma = 1/(1-\rho)$.

(a) Derive demand from the first-order conditions. Explain why the optimum is interior for $\rho < 1$ and why maximizing the sum of powers is incorrect when $\rho < 0$.

(b) At $p = (1, 4)$ and $w = 120$, calculate quantities and expenditure shares for $\sigma \in \{1/2, 1, 2\}$. Check budget exhaustion.

(c) Raise only the second price to 9. Recalculate the three demands and explain why a smaller quantity can have a larger expenditure share.

(d) Derive the own-price, cross-price, and wealth elasticities. Distinguish the elasticity of a quantity ratio from an individual demand elasticity.

Problem 1.4 [Core]. Substitutes, complements, and boundary choices.

Consider separately $u^A(x) = 2x_1 + x_2$ and $u^B(x) = \min\{x_1/3, x_2\}$ on $\mathbb{R}^2_+$, at strictly positive prices and wealth.

(a) Derive the full demand correspondence for u^A, including the price ratio at which it is set-valued.

(b) Derive demand for u^B. At $p = (2, 3)$ and $w = 36$, compute both models' choices.

(c) Explain why interior differentiable tangency conditions do not characterize these optima.

(d) Obtain the ratio $x_1 = 3x_2$ as a CES limit. Explain why holding the usual taste weights fixed does not preserve unequal Leontief proportions.

Part II: Formal and applied extensions

Problem 1.3 [Applied]. CES calibration and observational equivalence.

At prices $p^0 = (1, 2, 4)$, a household with wealth $w^0 = 120$ chooses $x^0 = (30, 25, 10)$. Consider Chapter 1's CES representation with positive normalized weights.

(a) For a specified $\sigma > 0$, derive the weights reproducing this observation.

(b) Calibrate the weights for $\sigma = 1/2$ and $\sigma = 2$. For each calibration predict demand at $p^1 = (2, 2, 4)$ and wealth 120.

(c) Explain why fitting one observation does not identify substitution. Describe additional observations that could distinguish the models.

Problem 1.6 [Applied]. KKT regimes and the quantity--quality tradeoff.

Unified Growth Theory links the demographic transition to a shift from child quantity toward education and child quality. This exercise isolates a stylized household block from Galor (2005); it is not the complete dynamic growth model. A parent with potential income $y > \bar{c}$ chooses consumption $c > 0$, fertility $n > 0$, and education time per child $e \geq 0$ to maximize $(1-\gamma)\log c + \gamma \log[nh_a(e)]$, where $h_a(e) = 1 + ae/(\tau + e)$, $0 < \gamma < 1$, $\tau > 0$ is baseline time per child, $a > 0$ indexes the return to education, and $\bar{c} > 0$ is subsistence consumption. The resource and subsistence constraints are $c + yn(\tau + e) \leq y$ and $c \geq \bar{c}$. The function h_a is a closed-form teaching specialization of Galor's general child-human-capital technology; the normalization makes one unit of e one unit of parental time.

Part I. By hand

(a) Write the complete KKT system in (c, n, e), including feasibility, multiplier signs, and complementary slackness. Prove that the resource constraint binds.

(b) Define total child-rearing time $t = n(\tau + e)$. Eliminate c and n, state the feasible set for (t, e), and show that the transformed objective is additively separable in t and e. Write its KKT system. Explain why an inequality can be active with a zero multiplier at a threshold.

(c) Solve globally for (t^*, e^*), then recover (c^*, n^*). Divide the answer into all parameter regions, including the equality cases. A stationary-point calculation alone is not sufficient. Hint: determine the sign of $h_a'(e)(\tau + e) - h_a(e)$.

(d) Let $\tau = 1/4$, $\gamma = 1/2$, and $\bar{c} = 4$. Calculate $(c^*, n^*, e^*, h_a(e^*))$ for every combination $a \in \{1/2, 3\}$ and $y \in \{6, 12\}$. Identify the active inequalities, calculate their transformed-problem multipliers, and verify feasibility and complementary slackness.

(e) Explain the income mechanism while subsistence binds, the quantity--quality response when a rises, and which dynamic elements of Unified Growth Theory this static exercise deliberately leaves out.

Part II. Verified scale-up with Codex and Python

(f) Record four lines before coding: Seed: the four hand calculations; Fixed: preferences, h_a, continuous fertility, and $(\tau, \gamma, \bar{c})$; Scaled: only the number of (a, y) configurations; Risk: threshold

misclassification, invalid parameter domains, or mistaking finite computation for proof.

(g) Create a file named work/ugt_kkt.py. On a deterministic rectangular grid containing $a = 1$ and $y = \overline{c}/(1-\gamma)$, compute the closed-form candidate and the transformed-problem KKT multipliers. Reject $a \leq 0$, $\tau \leq 0$, $\overline{c} \leq 0$, $\gamma \notin (0,1)$, and $y \leq \overline{c}$.

(h) At every grid point, check the resource identity, subsistence feasibility, nonnegativity, stationarity, complementary slackness, and agreement with an independently implemented bounded numerical maximization. State the numerical tolerances. Plot e^* and n^* and mark the two analytic threshold lines.

(i) Include tests immediately below, at, and above both thresholds. Report separately what the hand argument proves for every admissible parameter vector, what the program checks only on its finite grid, and what remains an empirical or dynamic modeling claim.

Household demand across subsistence and discretionary regimes

Chapter use case. Seed: Problems 1.1, 1.4. Fixed: all unlisted primitives and assumptions. Scaled: wealth from two hand calculations to a grid, with preferences and prices fixed. Tool: Codex + Python. Verification: Reproduce both hand wealth levels; classify infeasible, boundary, and interior regimes separately for each model; check feasibility, budget exhaustion, and continuity at every valid threshold.

Problem 1.5 [Applied]. Subsistence and quasilinear corners.

The first of two consumers has $u(x) = (x_1 - 1)^{1/3}(x_2 - 2)^{2/3}$ on $x \geq (1,2)$. The second has $u(x) = 4\sqrt{x_1} + x_2$ on $\mathbb{R}^2_+$.

(a) Derive the first consumer's demand above subsistence expenditure. State what happens at and below that threshold.

(b) At $p = (2,1)$, compute the first consumer's demands and shares at wealth 10 and 22. Distinguish homotheticity in discretionary consumption from homotheticity in total consumption.

(c) Normalize the second consumer's numeraire price to one and write $r = p_1$. Derive demand for all $r, w > 0$, including the zero-numeraire region.

(d) Identify where a zero wealth effect for the non-numeraire good is valid. Compare this restriction with the Stone--Geary model.

Chapter 2

Duality, Comparative Statics, and Welfare

Part I: By hand

Problem 2.1 [Core]. Cobb-Douglas duality.

For $u(x) = x_1^{\alpha} x_2^{1-\alpha}$ with $0 < \alpha < 1$, $p \gg 0$, $w > 0$, and $\overline{u} > 0$, derive the Marshallian demand $x(p, w)$, indirect utility $v(p, w)$, Hicksian demand $h(p, \overline{u})$, and expenditure function $e(p, \overline{u})$.

(a) Verify Walras' law and the identity $e(p, v(p, w)) = w$.

(b) Verify $h(p, v(p, w)) = x(p, w)$.

(c) Verify Roy's identity and Shephard's lemma.

Problem 2.5 [Core]. Translog shares and the regular price region.

Consider a homothetic translog index with $\alpha_0 = 0$, $\alpha = (2/5, 3/5)$, and $\Gamma = -\frac{1}{8}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)$.

(a) Derive shares and demands as functions of relative prices and wealth. Find the open interval of p_1/p_2 for which both shares are positive.

(b) Compute $H = \Gamma + \boldsymbol{b}\boldsymbol{b}^{\top} - \operatorname{diag}(\boldsymbol{b})$. Prove negative semidefiniteness throughout that interval.

(c) At $p = (2, 1)$ and $w = 100$, calculate demand, Marshallian elasticities, and compensated elasticities. Verify adding up and homogeneity.

(d) Explain why setting a negative predicted share to zero while retaining the other formula does not generally give a valid extension.

Part II: Formal and applied extensions

Problem 2.2 [Proof]. Differential restrictions on demand.

Let $x : \mathbb{R}^L_{++} \times \mathbb{R}_{++} \to \mathbb{R}^L_+$ be continuously differentiable and satisfy Walras' law.

(a) Differentiate Walras' law with respect to wealth and prices.

(b) Define the Slutsky matrix by $S(p, w) = D_p x(p, w) + D_w x(p, w) x(p, w)^{\top}$. Prove that $p^{\top} S(p, w) = 0$.

(c) Explain which additional restrictions on S follow from utility maximization and which of them do not follow from Walras' law alone.

Problem 2.3 [Proof]. The Slutsky matrix for Cobb-Douglas demand.

Using the Marshallian demand from Problem 2.1, compute $S(p, w) = D_p x + D_w x\, x^\top$.

(a) Show that S is symmetric.

(b) Show directly that $z^\top S z \leq 0$ for every $z \in \mathbb{R}^2$.

(c) Find the null space of S and interpret it using homogeneity.

Problem 2.4 [Proof]. CES duality and compensated substitution.

Use Chapter 1's CES utility, with positive normalized weights and $\sigma > 0$. For $\sigma \neq 1$, put $D(p) := \sum_k a_k^\sigma p_k^{1-\sigma}$.

(a) Derive $\mathcal{P}(p)$, expenditure, and Hicksian demand. Treat $\sigma = 1$ separately and retain the constants implied by the utility normalization.

(b) Show that the Slutsky matrix has off-diagonal entries $S_{\ell k} = \sigma x_\ell x_k / w$ and diagonal entries $S_{\ell\ell} = -\sigma x_\ell (1 - b_\ell)/p_\ell$. Prove symmetry and negative semidefiniteness using a weighted-variance argument. Identify its null space for strictly positive demand.

(c) Explain how compensated cross-price effects can be positive when uncompensated cross-price effects are negative.

(d) For equal weights, $\sigma = 2$, initial prices $(1, 3)$, wealth 90, and final prices $(2, 3)$, calculate compensation to retain initial utility. Decompose the first quantity's change into a compensated change and the remaining wealth effect.

Local CES and translog welfare approximation

Chapter use case. Seed: Problems 2.4, 2.5. Fixed: all unlisted primitives and assumptions. Scaled: log relative-price distance from one reference price. Tool: Codex + Python. Verification: Match the CES value, gradient, and Hessian at the reference price; enforce positive shares and negative semidefiniteness on the translog regular region; report welfare comparisons only on that region.

Problem 2.6 [Applied]. A translog approximation to CES demand.

Fix a two-good CES model, a positive reference price vector p^0, and substitution elasticity $\sigma > 0$. Write b_1^0, b_2^0 for the reference shares and use normalized prices $r_\ell = p_\ell / p_\ell^0$.

(a) Compute the gradient and Hessian of $\log \mathcal{P}_{\mathrm{CES}}$ with respect to log prices at p^0.

(b) Choose translog coefficients in $\log r$ so the log price index agrees with CES through second order at p^0.

(c) Verify the restrictions for homogeneity. Show that demand levels and local price responses agree at the reference observation.

(d) Explain why predictions can disagree for a large price change. State the regularity checks needed before making welfare comparisons.

Chapter 3

Revealed Preference and Recoverability

Part I: By hand

Problem 3.1 [Core]. A two-observation WARP test.

A consumer chooses $x^1 = (2, 2)$ at $(p^1, w^1) = ((2, 1), 6)$ and $x^2 = (1, 5/2)$ at $(p^2, w^2) = ((1, 2), 6)$.

(a) Verify that each observed bundle exhausts its budget.

(b) Determine whether each bundle was affordable at the other observation.

(c) Does the dataset satisfy WARP for single-valued demand? Identify the two direct revealed-preference comparisons that determine your answer.

Problem 3.2 [Core]. Recovering a quasilinear utility.

Normalize $p_2 = 1$ and suppose an interior demand system satisfies $x_1(p_1, w) = \beta - \alpha p_1$, where $\alpha, \beta > 0$, and Walras' law.

(a) Derive $x_2(p_1, w)$ and state the domain on which both demands are nonnegative.

(b) Compute the normalized scalar Slutsky term for the non-numeraire demand.

(c) Recover a quasilinear utility representation $u(x_1, x_2) = g(x_1) + x_2$ that generates the demand on the interior domain. Distinguish general ordinal uniqueness up to an increasing transformation from the narrower normalization freedom after the coefficient on x_2 is fixed at one.

Problem 3.3 [Core]. Afriat inequalities by hand.

Consider two observations: $p^1 = (1, 1)$, $x^1 = (2, 1)$ and $p^2 = (2, 1)$, $x^2 = (1, 1)$.

(a) Compute observed expenditures and all direct revealed-preference comparisons.

(b) Check GARP.

(c) Verify that $U^1 = 1$, $U^2 = 0$, and $\lambda^1 = \lambda^2 = 1$ satisfy the Afriat inequalities.

(d) Write the corresponding piecewise-linear Afriat utility and verify that it assigns the required utility levels at the observations.

Problem 3.5 [Core]. Classical utility as singleton CMU.

Let $u : X \to \mathbb{R}$ be continuous, let $\Omega_u = \{\{u\}\}$, and define $r_{\Omega_u}(x, y) = \max_{U \in \Omega_u} \min_{v \in U} [v(x) - v(y)]$.

(a) Prove that $r_{\Omega_u}(x, y) = u(x) - u(y)$.

(b) Prove that $x_{r_{\Omega_u}}(p, w) = \arg\max_{x \in B(p,w)} u(x)$.

(c) Explain precisely which conclusions require continuity, strict monotonicity, or concavity of u, and which do not.

Part II: Formal and applied extensions

Problem 3.6 [Proof]. WGARP without GARP.

Consider prices $p^1 = (4, 1, 5)$, $p^2 = (5, 4, 1)$, $p^3 = (1, 5, 4)$ and choices $x^1 = (4, 1, 1)$, $x^2 = (1, 4, 1)$, $x^3 = (1, 1, 4)$.

(a) Compute the 3×3 matrix with (t, s) entry $p^t \cdot x^s$.

(b) Show that the data satisfy WGARP and WARP but violate GARP. Identify the revealed three-cycle and explain why no pair forms a reversal.

(c) Suppose an asymmetric, strictly increasing preference function r rationalizes two choices $x \in x_r(p, w)$ and $x' \in x_r(p', p' \cdot x)$. Prove the compensated law of demand $(p' - p) \cdot (x' - x) \leq 0$. State where nonemptiness of demand is used.

A counterfactual demand support set

Chapter use case. Seed: Problems 3.1, 3.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of observed budgets from two to T, with two goods and the counterfactual budget fixed. Tool: Codex + Python; Lean for candidate certificates. Verification: Recover the hand support interval and verify that added observations can only shrink it. Lean may check GARP or Afriat certificates for candidates; it does not yet certify the full support-set characterization.

Problem 3.4 [Applied]. Demand forecasting with Varian's support set.

A finite dataset $\mathcal{O} = \{(p^t, x^t)\}_{t \in \mathbb{T}}$ satisfies GARP. Fix a counterfactual budget (p^0, w^0) with $p^0 \gg 0$, and define $\mathcal{S}_{\mathcal{O}}(p^0, w^0) = \{x^0 : p^0 \cdot x^0 = w^0,\ \mathcal{O} \cup \{(p^0, x^0)\} \text{ satisfies GARP}\}$.

(a) State a revealed-preference condition under which the augmented data imply $x^t \succeq^R x^0$.

(b) Derive the associated linear restriction on $p^0 \cdot x^0$.

(c) Explain why the restrictions in parts (a)--(b) are necessary but must be combined with every revealed-preference path into and out of the candidate observation to characterize $\mathcal{S}_{\mathcal{O}}(p^0, w^0)$. Explain why this set gives bounds rather than point identification.

Chapter 4

Choice Without Budget Geometry

Part I: By hand

Problem 4.1 [Core]. Incomplete preferences and multiple utilities.

Let $X = \{a, b, c\}$ and let R contain all reflexive pairs and the comparison aRb, but no other comparison between distinct alternatives.

(a) Show that R is a preorder but is incomplete.

(b) Prove that no real-valued utility function can represent R in the sense xRy if and only if $u(x) \geq u(y)$.

(c) Construct two utility functions u_1, u_2 such that xRy if and only if $u_k(x) \geq u_k(y)$ for both $k = 1, 2$.

Part II: Formal and applied extensions

Problem 4.3 [Proof]. Maximization and WGARP.

Let R be complete and transitive on a finite set X. For each nonempty menu $A \subseteq X$, let $c(A)$ be the set of R-greatest elements of A.

(a) Prove that $c(A)$ is nonempty.

(b) Suppose $x \in c(A)$, $y \in c(B)$, and $x, y \in A \cap B$. Prove that xI_Ry, where xI_Ry means xRy and yRx.

(c) Explain why this conclusion is WGARP, the appropriate weak axiom for a set-valued choice correspondence, and distinguish it from WARP for single-valued demand.

A menu-design rationality audit

Chapter use case. Seed: Problem 4.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of observed menus on the fixed set $X = \{a, b, c\}$. Tool: Codex + Lean. Verification: Recover the three-menu cycle and the one-observation repair; return an exact cycle certificate or

a complete transitive rationalization. The current pairwise-WARP theorem does not exclude this three-cycle; the general checker theorem is a new target.

Problem 4.2 [Core]. Choice cycles on finite menus.

Let $X = \{a, b, c\}$ and suppose a single-valued choice function satisfies $c(\{a, b\}) = a$, $c(\{b, c\}) = b$, and $c(\{a, c\}) = c$. Say that a preference relation R rationalizes these data when the observed alternative is the unique R-greatest element of each observed menu.

(a) Write the direct revealed-preference comparisons generated by these choices.

(b) Show that no transitive preference relation can rationalize the data.

(c) State the smallest change to one observation that restores acyclicity, and exhibit a complete and transitive rationalization of the revised data.

Part II

Risk, Stochastic Choice, and Attention

Chapter 5

Expected Utility and Risk

Part I: By hand

Problem 5.1 [Core]. Affine expected-utility representations.

Let $Z = \{z_1, \ldots, z_n\}$ and suppose $U(\mu) = \sum_j \mu_j u(z_j)$ represents a nontrivial preference on $\Delta(Z)$.

(a) Show directly that U is affine in lotteries.

(b) Prove that $U' = aU + b$ with $a > 0$ represents the same preference.

(c) Show that if two expected-utility representations agree on preferences, their Bernoulli indices differ by a positive affine transformation.

Problem 5.2 [Core]. First-order stochastic dominance.

Lotteries μ and ν on prizes $\{0, 1, 2\}$ have probabilities $\mu = (0.2, 0.3, 0.5)$ and $\nu = (0.3, 0.4, 0.3)$.

(a) Compare their cumulative distribution functions and determine which lottery first-order stochastically dominates the other.

(b) Prove the expected-utility ranking for every increasing Bernoulli utility by discrete summation by parts.

(c) Explain why strict dominance need not imply a strict preference for a weakly increasing utility.

Part II: Formal and applied extensions

Problem 5.3 [Proof]. The Arrow-Pratt approximation.

Let $\varepsilon_\sigma = \sigma\xi$, where $\mathbb{E}\xi = 0$, $\mathbb{E}\xi^2 = 1$, and ξ is bounded. An agent with wealth w has utility u that is twice continuously differentiable near w, with $u'(w) > 0$. For all sufficiently small $\sigma \geq 0$, local invertibility gives a unique solution near zero to the indifference equation below. Define the monetary risk premium $\mathrm{RP}(\sigma)$ to be that local branch, characterized by $\mathrm{RP}(\sigma) \to 0$ as $\sigma \downarrow 0$:

$u(w - \mathrm{RP}(\sigma)) = \mathbb{E}[u(w + \varepsilon_\sigma)]$.

(a) Use second-order Taylor expansions to derive the approximation

$\mathrm{RP}(\sigma) = \frac{1}{2}A(w)\sigma^2 + o(\sigma^2)$,

where $A(w) = -u''(w)/u'(w)$.

(b) State the regularity and small-risk qualifications needed for this approximation.

(c) Explain why the formula is invariant to positive affine transformations of u.

A multi-asset portfolio under risk

Chapter use case. Seed: Problems 5.3, 5.4. Fixed: all unlisted primitives and assumptions. Scaled: the number of jointly normal risky assets from one to K, retaining CARA utility and unconstrained positions. Tool: Codex + Python. Verification: Recover the one-asset formula; verify the first-order condition and concavity using the covariance matrix; report singular-covariance cases explicitly and do not add portfolio constraints silently.

Problem 5.4 [Applied]. CARA-normal portfolio demand.

Terminal wealth is $W_T = wR_f + \theta(\widetilde{R} - R_f)$, where $\widetilde{R} \sim N(m_R, \sigma^2)$, R_f is risk free, $\theta \in \mathbb{R}$ (borrowing and short selling are unrestricted), and utility is $u(W_T) = -\exp(-aW_T)$ with $a > 0$.

(a) Show that maximizing expected utility is equivalent to maximizing a certainty equivalent.

(b) Derive optimal risky investment θ^*.

(c) Give comparative statics with respect to $m_R - R_f$, a, and σ^2.

Chapter 6

Stochastic Choice and Random Utility

Part I: By hand

Problem 6.1 [Core]. Logit odds and IIA.

Choice probabilities satisfy $P(j \mid A) = \exp(\delta_j/\tau)/\sum_{k\in A}\exp(\delta_k/\tau)$, where $\tau > 0$ and δ_j is deterministic net surplus.

(a) Derive the odds ratio $P(j \mid A)/P(k \mid A)$.

(b) Prove the independence-of-irrelevant-alternatives property.

(c) Explain precisely which part of IIA is an empirical restriction and which part is an algebraic implication of the model.

Problem 6.3 [Core]. A finite random-utility mixture.

There are alternatives a, b, c. A population consists of three strict preference types: $a \succ b \succ c$ with mass 0.5, $c \succ a \succ b$ with mass 0.2, and $b \succ a \succ c$ with mass 0.3.

(a) Compute choice probabilities from menus $\{a, b\}$ and $\{a, b, c\}$.

(b) Let the columns of M correspond to all six strict orders on $\{a, b, c\}$, with zero masses allowed. Stack the observed probabilities as $\mathbf{P}$ and write $\mathbf{P} = M\lambda$.

(c) Explain why nonnegativity of λ and $\mathbf{1}_6^\top \lambda = 1$ are the central finite random-utility feasibility restrictions.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

Red bus, blue bus, and duplicated products

Chapter use case. Seed: Problems 6.1, 6.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of observationally identical bus labels under independent-logit shocks. Tool: Codex + Python; Lean for the IIA core. Verification: Recover the one-bus and two-bus

probabilities, verify that probabilities sum to one, and separate the verified logit odds result from any shared-shock comparison model.

Problem 6.2 [Applied]. The red-bus blue-bus problem.

Walking and a red bus initially have systematic utilities normalized so that each has probability one half under logit. A blue bus with the same systematic utility as the red bus is added.

(a) Compute the new logit probabilities.

(b) Compare the total bus share before and after entry.

(c) Explain why a model with highly correlated bus-specific shocks would predict a different substitution pattern.

Chapter 7

Limited Attention and Costly Information

Part I: By hand

Problem 7.1 [Core]. Sparse-max quasilinear demand.

Let $u(x_1, x_2) = \sqrt{x_1} + x_2$. The consumer chooses $x \in \mathbb{R}^2_+$, uses perceived prices $\widehat{p} \gg 0$ in the first-order condition, and must satisfy the actual budget $p \cdot x = w$, where $p \gg 0$ and $w > 0$.

(a) Derive the interior sparse-max choice.

(b) State the condition under which the implied x_2 is nonnegative.

(c) Show that ordinary Marshallian demand is recovered when $\widehat{p} = p$, including the corner when the interior condition fails.

Problem 7.2 [Core]. CES demand under perceived prices.

Let $u(x) = [\alpha x_1^\rho + (1-\alpha)x_2^\rho]^{1/\rho}$, where $0 < \alpha < 1$ and $\rho < 1$, $\rho \neq 0$. Consider an interior choice $x \in \mathbb{R}^2_{++}$ with perceived prices $\widehat{p} \gg 0$, actual prices $p \gg 0$, and wealth $w > 0$.

(a) Use perceived prices $\widehat{p}$ to derive the chosen ratio x_1/x_2.

(b) Scale the bundle so that it exhausts the actual budget $p \cdot x = w$.

(c) Identify separately the perceived-price composition effect of changing $\widehat{p}$ and the feasibility effect of changing p. Explain why the first is not automatically a Hicksian substitution effect.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

Paying for information before choosing

Chapter use case. Seed: Problem 7.3. Fixed: all unlisted primitives and assumptions. Scaled: the common number N of equally likely states and matching actions, beginning at $N = 2$. Tool: Codex + Python. Verification: Recover the binary optimum, enforce conditional-probability and information-cost identities, and label any symmetry restriction used in the N-state extension. Problems 7.1 and 7.2 are comparison models, not derivations of rational inattention.

Problem 7.3 [Applied]. A binary rational-inattention problem.

There are two equally likely states and two actions. The decision maker receives payoff one from the correct action and zero from the incorrect action. A symmetric strategy is correct with probability $q \in [1/2, 1]$ and pays Shannon-information cost $\kappa[\log 2 - H(q)]$, where $H(q) = -q \log q - (1-q)\log(1-q)$ and $\kappa > 0$.

(a) Write expected net payoff as a function of q.

(b) Derive the interior optimum and show that $q^* = \exp(1/\kappa)/(1 + \exp(1/\kappa))$.

(c) Describe the limits as $\kappa \downarrow 0$ and $\kappa \uparrow \infty$.

Part III

Aggregation and Markets

Chapter 8

Aggregate Demand and Representative Consumers

Part I: By hand

Problem 8.1 [Core]. Gorman aggregation.

Suppose consumer i has indirect utility $v_i(p, w_i) = a_i(p) + b(p)w_i$, with the same positive function b for every consumer. Assume a_i and b are differentiable on the relevant positive-price domain and that each indirect utility satisfies the regularity conditions for Roy's identity.

(a) Apply Roy's identity to derive individual demand.

(b) Show that aggregate demand depends on the wealth distribution only through total wealth $W = \sum_i w_i$.

(c) Construct a representative indirect utility that generates aggregate demand.

Problem 8.2 [Core]. Aggregation under common homothetic preferences.

All consumers have the same homothetic preferences, strictly positive wealth $w_i > 0$, and a single-valued demand. Normalize the unit-wealth demand as $x(p, 1)$.

(a) Show that $x_i(p, w_i) = w_i x(p, 1)$.

(b) Derive aggregate demand and identify a representative consumer.

(c) Explain why common homotheticity is sufficient but not necessary for exact aggregation.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

Aggregation when household wealth moves with prices

Chapter use case. Seed: Problems 8.1, 8.2. Fixed: all unlisted primitives and assumptions. Scaled: the number of consumers, with each wealth rule and preference class fixed. Tool: Codex + Python; Lean for the finite-sum core. Verification: Verify the finite aggregation identity and isolate every wealth-revaluation term. Do not describe the derivative or Slutsky bridge as Lean verified.

Problem 8.3 [Proof]. Why uncompensated aggregate price derivatives need not be symmetric.

Let $x^A(p) = \sum_i x_i(p, w_i(p))$, where individual wealth may vary with prices through endowments or profit shares.

(a) Differentiate x^A with respect to prices.

(b) Separate compensated substitution terms from wealth-revaluation terms.

(c) Explain why symmetry and negative semidefiniteness of each individual Slutsky matrix do not generally imply the same properties for the uncompensated derivative $D_p x^A$.

Part IV

Production and Equilibrium

Chapter 9

Theory of the Firm

Part I: By hand

Problem 9.1 [Core]. Cobb-Douglas production and cost.

A firm has production function $f(\boldsymbol{z}) = z_1^{\alpha} z_2^{\beta}$ with $\alpha, \beta > 0$.

(a) Classify returns to scale as a function of $\alpha + \beta$.

(b) Derive conditional factor demands and the cost function for output $q > 0$ and input prices $\boldsymbol{w} \gg 0$.

(c) Verify that cost is homogeneous of degree one and concave in $\boldsymbol{w}$.

Problem 9.3 [Core]. Constant returns and unbounded profit.

Assume Y is a cone and $0 \in Y$.

(a) Prove that either $\pi_Y(p) = 0$ or $\pi_Y(p) = +\infty$.

(b) Characterize the price vectors for which profit is zero.

(c) Explain why a competitive constant-returns firm can operate at any scale only when the zero-profit condition holds.

Advanced exam practice

The suggested time applies to the hand questions, not the optional tool extension. Start with a diagram or solution plan, state the constraints, check all regimes and explain global optimality and economic accounting. Attempt the hand questions before using tools. These problems are additional preparation, not extra weekly requirements unless assigned in class. Before an optional extension, state Seed, Fixed, Scaled and Risk and enlarge one feature at a time.

Difficulty guide. Problems 9.A1 to 9.A4 provide new optimization and regime practice. Problem 9.A5 restores the Western 2017 final examination question on indivisible production, including its original endowments and global equilibrium reasoning. Problem 9.A6 adds a simple housing production case connecting developer optimization, latent prices, land rents and general equilibrium.

Problem 9.A1 [Advanced]. Automation, production lines and the value of capacity.

Suggested time: 35 minutes. Total: 30 points. The optional tool extension is not included in the suggested time.

A competitive manufacturer sells a homogeneous output at the fixed price $p = 8$. It operates two production lines. Their outputs are $q_1, q_2 \geq 0$, and their combined capacity is $q_1 + q_2 \leq 6$. Costs are $C_1(q_1) = 2q_1 + \frac{1}{2}q_1^2, \qquad C_2(q_2; a) = aq_2 + \frac{1}{2}q_2^2$, where $a \geq 0$ is a technology parameter, not a choice. The manufacturer maximizes revenue less both production costs. There are no fixed costs.

(a) Write the optimization problem and its KKT conditions. Use a nonnegative multiplier λ for total capacity. Explain how the condition for a line with zero output differs from the equality for a line with positive output. [5]

(b) For $a = 4$, determine both outputs, profit and λ. Check every KKT condition rather than only solving the interior equations. [5]

(c) An automation improvement reduces a from 4 to 1, without changing output price or capacity. Solve the new problem. Explain why output on the first line can fall even though total technology has improved. [5]

(d) Keeping capacity equal to 6, solve for all $a \geq 0$. Identify the value at which the second line shuts down. At that value, distinguish a binding capacity constraint from a strictly positive shadow value. [5]

(e) Prove global optimality and uniqueness of your proposed allocation. You may use strict concavity and KKT sufficiency, but state the premises. Alternatively give an exact expression for the profit loss from any feasible deviation. [5]

(f) Let $V(K, a)$ be maximum profit when total capacity is K. Interpret $\partial V/\partial K$ at $K = 6$ for $a = 4$ and $a = 1$. Explain why the automation improvement raises the value of capacity in this example. [5]

Optional tool extension. Extend the manufacturer to ten lines with costs $c_j q_j + \kappa_j q_j^2/2$, where $\kappa_j > 0$. Sort the entry thresholds, solve the capacity equation, and test primal feasibility, multiplier signs, complementary slackness and every excluded line. Compare direct optimization with the exact active set method. A numerical solver is a check, not the proof requested in part (e).

Problem 9.A2 [Advanced]. Human input, machine capacity and cost duality.

Suggested time: 45 minutes. Total: 36 points. The optional tool extension is not included in the suggested time.

A firm uses labor $\ell \geq 0$ and machine services $r \geq 0$. It can produce output $q \geq 0$ satisfying $q \leq \sqrt{\ell} + \sqrt{ar}, \qquad r \leq R$. Here $a > 0$ is machine productivity and $R > 0$ is the available quantity of machine services. Labor and machine prices are $w > 0$ and $v > 0$. The firm pays $w\ell + vr$. All inputs and outputs are continuously divisible, and there is no fixed cost.

(a) For a required output $q > 0$, write the cost minimization problem and show that its production constraint binds. Using $x = \sqrt{\ell}$ and $y = \sqrt{ar}$, reduce it to a strictly convex quadratic allocation problem. [5]

(b) Find the unique conditional input demands and cost function $C(w, v, q; a, R)$ for every $q > 0$. Identify the output threshold at which the machine capacity starts binding, and verify that cost and marginal cost match across the threshold. [8]

(c) At a fixed output price $p > 0$, derive optimal labor, machine use, output and profit. Give the complete price partition, including the boundary. [7]

(d) Verify Hotelling's lemma with respect to p in both regimes. Verify Shephard's lemma with respect to w and v in the conditional cost problem, stating which objects are held fixed. [6]

(e) Take $p = 3$, $w = v = 1$ and $R = 1$. Compare $a = 1$ with $a = 4$. Calculate output, both inputs, profit and the machine capacity multiplier in each case. Explain why greater productivity need not make the capacity constraint less valuable. [5]

(f) Explain why the proposed solution is globally optimal. Distinguish improving machine productivity a from increasing available machine services R. Which model feature prevents the firm from using an unlimited amount of the improved technology? [5]

Optional tool extension. Draw the two dimensional regime map in (p, R) for several values of a. Check continuity and envelope identities at the boundary using exact formulas before numerical differentiation. Report the analytical statement separately from any graph or solver output.

Problem 9.A3 [Advanced]. A common emissions budget and the limits of firm aggregation.

Suggested time: 40 minutes. Total: 32 points. The optional tool extension is not included in the suggested time.

Two production units sell the same output at the fixed competitive price $p = 6$. Their costs are $C_1(q_1) = q_1 + \frac{1}{2}q_1^2$, $C_2(q_2) = 2q_2 + \frac{1}{2}q_2^2$. Outputs satisfy $q_1, q_2 \geq 0$. Initially each unit of output from unit 1 uses one emissions permit and each unit from unit 2 uses two. A manufacturer owns both units and has a total emissions budget $B > 0$, so $q_1 + 2q_2 \leq B$. It maximizes total revenue less production costs. There are no other constraints or fixed costs.

(a) Derive the KKT system and explain the units of the common constraint multiplier. Why is adding the two unconstrained supplies generally incorrect? [5]

(b) Solve the problem for every $B > 0$, finding all thresholds at which a unit enters or the emissions budget ceases to have a positive shadow value. Check the boundary values. [8]

(c) At $B = 4$, compute outputs, gross revenue less production costs and the emissions multiplier. Prove that the allocation is globally optimal and unique. [5]

(d) Suppose instead that independent firms operate the units and permits can be traded at the competitive price $\tau \geq 0$. Each firm pays τ times its emissions. Show that an appropriate permit price implements the allocation from part (c). Explain why the initial ownership of permits affects income but not these output choices under the stated assumptions. [5]

(e) An improvement reduces the emissions requirement of unit 2 from two permits to one, holding $p = 6$ and $B = 4$ fixed. Solve the new problem and compare each output, their total, production profit before permit payments, and the permit price. Explain why the price of the scarce permit can rise when a production unit becomes cleaner. [6]

(f) Explain the difference between this manufacturer problem and a social planner problem that values consumer surplus or environmental damage. State one missing primitive needed to make a welfare claim. [3]

Optional tool extension. Add heterogeneous units with costs $c_j q_j + q_j^2/(2e_j)$ and emissions coefficients $b_j > 0$. Construct $q_j(\tau) = e_j[p - c_j - \tau b_j]_+$ and solve the permit clearing equation. Check the entry inequalities and distinguish output changes from transfers of permit rents.

Problem 9.A4 [Advanced]. A fixed software cost, entry and a missing competitive equilibrium.

Suggested time: 45 minutes. Total: 36 points. The optional tool extension is not included in the suggested time.

A firm must activate a production system before producing. Activation is $d \in \{0, 1\}$ and output satisfies $0 \leq q \leq Kd$, where $K > 0$. Its profit at the fixed output price $p \geq 0$ is $pq - Fd - cq - \frac{q^2}{2e}$, where $F > 0$ is the activation cost, $c \geq 0$ is unit operating cost and $e > 0$ is productivity. Inactivity has $d = 0$, $q = 0$ and zero profit. All parameters are fixed when the firm chooses (d, q).

(a) Conditional on $d = 1$, solve for output using KKT, including the lower and upper output constraints. Why is this conditional calculation not enough to find a global optimum of the original problem? [6]

(b) Derive the entry price, allowing for the possibility that capacity is already binding at entry. State the entire output supply correspondence below, at and above that price. Do not discard tied optima. [9]

(c) Set $c = 2$, $e = 1$, $K = 4$ and $F = 2$. Calculate supply and maximum profit at $p = 3$, $p = 4$ and $p = 5$. Explain the jump in supply. [5]

(d) At the fixed price $p = 7/2$, an AI software improvement reduces F from 2 to $1/2$, without changing c, e or K. Compare activity, output and profit before and after the change. For prices at which the firm is active under both costs, does reducing F change its conditional output? [5]

(e) Return to $F = 2$. Suppose this firm is the only producer, consumers have aggregate demand $D(p) = \max\{5 - p, 0\}$, and all agents are price takers. Prove that there is no competitive market clearing price. Does the reduction to $F = 1/2$ restore an equilibrium? If so, find it and verify that the firm prefers to operate. [7]

(f) Explain which convexity property fails and why a smooth first order condition cannot rule out the failure in part (e). Distinguish the claim that no competitive equilibrium exists from the claim that no feasible or optimal allocation exists. [4]

Optional tool extension. For a population of such firms, vary the distribution of fixed costs and capacities. Track the supply correspondence, allowing for ties at entry. Compare an atomic firm population with a continuum. A plot that crosses interpolated supply points is not proof of exact market clearing.

Problem 9.A5 [Advanced]. Indivisible production and two endowment configurations.

Restored from the Western University Fall 2017 final examination. The indivisible technology, literal utility and two original endowment configurations are retained.

Suggested time: 40 minutes. Total: 40 points.

There is one consumer and one firm, with one physical input and one output good. The consumer owns the firm. A consumer bundle is $(c, d) \in \mathbb{R}^2_+$, where c is retained input and d is consumption of the output. Utility is $u(c, d) = 2 - c + 2d$. Thus the utility of retained input is decreasing, exactly as in the original displayed formula; do not replace it by a leisure utility.

The firm has the production set $Y = \bigcup_{a \in \mathbb{Z}_+} \{(-a, a) - v : v \in \mathbb{R}^2_+\}$, where $\mathbb{Z}_+ = \{0, 1, 2, \ldots\}$. It can choose an integer activity a that uses a units of input and produces a units of output, and it can freely dispose of either commodity. Normalize the output price to 1 and denote the input price by $r \geq 0$. The firm maximizes $(r, 1) \cdot y$ over $y \in Y$. If this maximum is finite and attained, denote it by $\pi(r)$.

For endowment $\omega = (\omega_c, \omega_d)$, the consumer budget is $rc + d \leq r\omega_c + \omega_d + \pi(r)$. A competitive equilibrium requires both global optimization problems and exact clearing $(c, d) = \omega + y$.

(a) Characterize the firm's profit and its complete optimizing production correspondence for all $r \geq 0$. Explain why free disposal and integer activity matter, and why the firm must not insert the economy's total input endowment as an additional private constraint. [12]

(b) Begin with the original endowment $\omega = (2, 0)$. Find every competitive equilibrium price, consumer bundle and production plan. Prove that no other price or integer activity can be an equilibrium. [12]

(c) Change only the endowment to the other original case, $\omega = (0, 2)$. Find the full equilibrium set and compare the input price determination with part (b). [10]

(d) Exhibit two plans in Y whose midpoint is not in Y. Explain why zero equilibrium profit does not make this technology a convex cone, and why first order calculus in a cannot replace the global production comparison. [6]

Problem 9.A6 [Advanced]. Housing values, latent prices and land rents.

This is an original teaching specialization of the housing production and value proxy ideas in Epple, Gordon and Sieg (2010) and Aguiar, Kashaev and Allen (2023). It is not a reproduction of either paper's general identification theorem.

Part I. By hand

Suggested time: 50 minutes. Total: 50 points. The optional tool extension is not included in the suggested time.

Housing services are a homogeneous output. Developers use land $L \geq 0$ and materials $M \geq 0$ to produce $H = 2A\sqrt{LM}$, where $A > 0$ is a common technology parameter. Materials are units of the numeraire and cost 1 each. The price of one unit of housing is $p_q > 0$ and the rent of one unit of land is $\rho > 0$. Every developer takes both prices as given and chooses its own scale. There is no fixed cost, market power or uncertainty. All prices and revenues refer to the same single period.

For a developer using positive land, define materials per unit of land $m = M/L$, housing per unit of land $q = H/L$, and observed housing revenue per unit of land $v = p_q q$. The conditional operating value per unit of land, before paying for land, is $\Pi(p_q; A) := \max_{m \geq 0}\{2Ap_q\sqrt{m} - m\}$. Net developer profit also subtracts ρL.

(a) Find the unique conditional materials intensity, housing intensity, revenue and operating value. Provide a global argument, using either KKT after a change of variable or an exact algebraic bound. Explain why a land payment that is fixed conditional on L does not change the choice of m. [8]

(b) Consider otherwise comparable markets with the baseline technology $A = 1$. Housing quantity q and its price p_q are not separately observed. The analyst observes v and land rent $\rho(v) = v/2$ for every $v > 0$, maintains zero net developer profit, and represents the latent price by $p_q = g(v)$. The units of housing are anchored by $g(2) = 1$. Use the conditional operating value from part (a), the observed rent and zero profit to recover the price, housing intensity and materials intensity by algebra. Check the anchor and explain why this parametric inversion does not identify an unrestricted unknown technology. [8]

(c) Return to any $A > 0$ and let a developer choose both L and M. Characterize its maximum net profit and all optimal input plans according to the relation between ρ and the conditional

operating value from part (a). Which relation permits positive land use and free entry without unbounded profit? Give a global certificate, not only two first order equalities. [6]

(d) Close a genuine general equilibrium. A single household has utility $u(C, H) = C + B \log H$, where $C \geq 0$ is numeraire consumption, $H > 0$ is housing and $B > 0$ is fixed. It owns a numeraire endowment $W > B$, all developers, and the fixed total land $\overline{L} > 0$. It receives land rent, and firms have zero net profit in a free entry equilibrium. Its budget is $C + p_q H \leq W + \rho\overline{L}$. Materials used by developers come from the numeraire endowment. Write household KKT conditions taking prices as given. Derive equilibrium prices and aggregate H, M, C. Verify household global optimality, developer global optimality, land clearing, housing clearing and the numeraire resource identity $C + M = W$. Distinguish uniqueness of aggregate equilibrium from the division of production among firms. [12]

(e) Set $W = 10$, $B = 4$, $\overline{L} = 2$ and initially $A = 1$. Calculate all prices, aggregate quantities and household income. Now double A without changing preferences or endowments. Calculate the new equilibrium and the change in v. Explain how positive land rents can coexist with zero developer profit. [8]

(f) For general $W > B$ and $\overline{L} > 0$, compare A and $2A$. Compute the household's utility gain and distinguish changes in physical housing from changes in housing expenditure. Can the analyst reuse the baseline price proxy and its anchor from part (b) after the technology changes? Explain what stays fixed in a valid comparison across markets and what fails when technology moves. State one reason this single location model does not by itself describe the spatial structure of a city. [8]

Part II. Codex and verification

Optional analytic bridge. Now regard the common production function per unit land as unknown rather than impose the stated Cobb Douglas form. Maintain interior conditional optima, differentiability, material price 1, the full observed rent curve $\rho(v) = v/2$, zero net profit and the anchor $g(2) = 1$. Use the envelope theorem and $v = p_q q$ to derive a differential equation for g. Recover the implied price and production relationship and explain which assumptions make this different from the algebraic inversion in part (b).

Seed: the two equilibria in part (e). Fixed: $W = 10$, $B = 4$, the technology class, ownership, material price normalization, preferences, and the strictly positive domains. Scaled: aggregate land $\overline{L} > 0$ over a grid, first at $A = 1$ and then under the same stated doubling of A. This changes one market feature at a time while repeating the fixed technology comparison. Risk: confusing a conditional land choice with fixed aggregate land, counting rent as an additional resource cost, or using the old price proxy outside its technology regime.

Build an equilibrium verifier rather than only plotting formulas. Check household KKT, the developer profit certificate, all three market identities, income accounting and the welfare difference. Reject invalid inputs. Include a test that deliberately reuses the old price proxy after the technology change and explain its failure. An optional Lean extension may certify the algebraic profit bound and the finite accounting identities from explicitly supplied positive price and equilibrium equations. Do not describe the empirical price proxy interpretation or a general existence theorem as Lean verified.

Part II: Formal and applied extensions

Problem 9.2 [Proof]. Dual properties of the profit function.

Let $Y \subseteq \mathbb{R}^L$ be nonempty, closed, and convex. Assume that profit $\pi_Y(p) = \sup_{y \in Y} p \cdot y$ is finite on a neighborhood of the price under study and that the supremum is attained.

(a) Prove that π_Y is convex and homogeneous of degree one in prices.

(b) If supply is unique at p, use a subgradient argument to prove Hotelling's lemma and explain why the profit function is differentiable there.

(c) State what changes when supply is set-valued.

Input substitution in a multi-input firm

Chapter use case. Seed: Problem 9.1. Fixed: all unlisted primitives and assumptions. Scaled: the input dimension from two to L, with the output target and technology family fixed. Tool: Codex + Python; Lean for finite algebra. Verification: Recover the two-input cost benchmark; check Shephard derivatives, Hessian symmetry, negative semidefiniteness, and the homogeneity null direction; state the analytic assumptions not encoded in Lean.

Problem 9.4 [Proof]. Shephard's lemma and substitution.

Suppose the minimum in $C(\boldsymbol{w}, q) = \min\{\boldsymbol{w} \cdot \boldsymbol{z} : f(\boldsymbol{z}) \geq q, \boldsymbol{z} \geq 0\}$ exists, conditional factor demand $\mathcal{Z}(\boldsymbol{w}, q)$ is single-valued, and C is twice differentiable in $\boldsymbol{w}$.

(a) Prove Shephard's lemma.

(b) Show that $D_{\boldsymbol{w}}\boldsymbol{z} = D^2_{\boldsymbol{ww}}C$ is symmetric and negative semidefinite.

(c) Use homogeneity to show $D_{\boldsymbol{w}}\boldsymbol{z}(\boldsymbol{w}, q)\boldsymbol{w} = 0$.

Chapter 10

Competitive Markets and Partial Equilibrium

Part I: By hand

Problem 10.1 [Core]. Linear partial equilibrium.

Demand and supply are $Q^d(p) = a - bp$ and $Q^s(p) = d(p - c)$ on their economically relevant ranges, with $a, b, d > 0$, $c \geq 0$, and $a > bc$.

(a) Solve for equilibrium price and quantity.

(b) Compute consumer and producer surplus using the choke price a/b of inverse demand and the price intercept c of inverse supply, so that the usual triangular-surplus geometry applies.

(c) Verify that the competitive quantity maximizes total surplus.

Advanced exam practice

The suggested time applies to the hand questions, not the optional tool extension. Start with a diagram or solution plan, state the constraints, check all regimes and explain global optimality and economic accounting. Attempt the hand questions before using tools. These problems are additional preparation, not extra weekly requirements unless assigned in class. Before an optional extension, state Seed, Fixed, Scaled and Risk and enlarge one feature at a time.

Difficulty guide. Problems 10.A1 to 10.A4 are preparatory practice. Problems 10.A5 and 10.A6 restore the harder continuum questions from the Western 2019 final examination and the SFU 2024 Problem Set 7. Their source notes distinguish necessary corrections from new extensions. Use the restored questions as the main timed exam preparation. Problem 10.A7 restores the historical equilibrium and demand estimation exercise, with explicit corrections and a labelled time series extension. Problem 10.A8 connects discrete choice demand to pricing and endogeneity before a BLP style computational extension.

Problem 10.A1 [Advanced]. Dispatch, capacity rents, and a technology improvement.

Suggested time: 40 minutes. Total: 40 points.

A competitive electricity market lasts one period. A representative buyer chooses electricity $Q \geq 0$ and money $c \geq 0$ to maximize $c + 14Q - Q^2/2$ subject to $c + pQ \leq 100$, where $p \geq 0$ is the electricity price. The buyer takes prices as given. There are two price taking suppliers. Supplier 1 chooses $q_1 \in [0, 4]$ and pays $C_1(q_1) = 4q_1 + q_1^2/2$. Supplier 2 chooses $q_2 \geq 0$ and initially pays $C_2(q_2) = 8q_2 + q_2^2$. Both suppliers can choose zero output at zero cost. Equilibrium requires $Q = q_1 + q_2$. Ignore externalities. All money payments are transfers between the buyer and the suppliers, and total surplus counts electricity benefits less real production costs.

(a) Derive the buyer's demand and both suppliers' supply functions for every $p \geq 0$. For supplier 1, write a KKT system including the capacity constraint and identify the price at which capacity begins to bind. Explain why each proposed choice is globally optimal. [10]

(b) Find the initial competitive equilibrium. Report p, Q, q_1, q_2, and the capacity multiplier of supplier 1. Check market clearing and the inequalities implied by your proposed regimes. [8]

(c) A software improvement changes only supplier 2's cost to $C_2^A(q_2) = 2q_2 + q_2^2$. There is no adoption cost. Find the new equilibrium and determine whether supplier 1's capacity still binds. Do not retain the old regime without checking it. [10]

(d) Calculate consumer surplus, each supplier's profit, and total surplus before and after the improvement. Does the improvement benefit every market participant? Explain why a supplier can lose profit even when total surplus rises. Interpret supplier 1's capacity multiplier in both equilibria as the local marginal social value of additional capacity, holding the technology regime fixed. [12]

Optional tool extension. Replace the capacity 4 by $K > 0$ and supplier 2's linear cost by $b \geq 0$. Use a script to enumerate candidate active sets, solve their market clearing equations, and reject candidates that violate output or capacity inequalities. Test the original and improved cases exactly using rational arithmetic. Plot equilibrium price, outputs, and the capacity value over a parameter grid. A graph is a diagnostic, not a proof of the regime boundaries.

Problem 10.A2 [Advanced]. A corrective tax that changes who participates.

Suggested time: 45 minutes. Total: 40 points.

Two consumers $i \in \{A, B\}$ choose $x_i \geq 0$ and money $c_i \geq 0$. Their utilities are $u_A = c_A + 12x_A - x_A^2/2$ and $u_B = c_B + 8x_B - x_B^2/2$, and each initially has 100 units of money. The consumer price is $p \geq 0$. A competitive producer chooses $Q \geq 0$ with cost $C(Q) = 2Q + Q^2/2$. A unit tax $t \in [0, 6]$ creates the wedge between the consumer price p and the producer's receipt $p - t$. The government returns the revenue as lump sum money transfers, which are not conditional on a consumer's own purchase. Wealth is sufficient that money remains positive in every case considered. Total purchases $Q = x_A + x_B$ generate real external damage $D(Q) = Q^2/2$, measured in money and imposed outside these two consumers' choice problems. Social welfare is their gross consumption benefit less production cost and damage. There are no other market failures.

(a) Derive individual demands and producer supply. Find the equilibrium at $t = 0$ and verify whether both consumers buy positive quantities. [8]

(b) Find equilibrium price and quantities as functions of t on $[0, 6]$. Identify all tax thresholds at which a consumer's choice changes regime. Then calculate the equilibrium at $t = 4$. Check the omitted consumer's entry inequality whenever you claim a zero purchase. [12]

(c) Formulate the planner's maximization over $x_A, x_B \geq 0$ and find the unique efficient allocation. Find a unit tax that implements it. Explain why simply assuming that both consumers remain active can give an incorrect tax. [12]

(d) Calculate social welfare at $t = 0$, $t = 4$, and at your efficient tax. For $t = 4$, reconcile welfare with consumer surplus, producer profit, government revenue, and external damage. Is it possible here for a welfare improving tax to reduce one consumer's market consumer surplus? Distinguish that surplus from the consumer's final welfare after any revenue transfer. [8]

Optional tool extension. Replace damage by $\delta Q^2/2$, where $\delta \geq 0$, and for this extension allow any unit tax $t \geq 0$ rather than retaining the upper bound 6. Enumerate the efficient active sets and derive the value of δ at which consumer B exits. Write exact tests for primal feasibility, all zero purchase inequalities, and the equality $t = \delta Q$ at the proposed implementing allocation. Numerical optimization can check examples but does not replace the regime proof.

Problem 10.A3 [Advanced]. Free entry, smaller plants, and the limit of marginal conditions.

Suggested time: 40 minutes. Total: 40 points.

A market has demand $D(p) = \max\{a - p, 0\}$, with $a > 6$. It is generated by a buyer who chooses $c, Q \geq 0$ to maximize $c + aQ - Q^2/2$ subject to $c + pQ \leq M$, where $p \geq 0$ and money wealth $M > a^2/4$. Each potential price taking plant may stay closed and pay zero, or open, choose $q > 0$, and pay $C_F(q) = F + q^2$, where $F > 0$ is an avoidable real setup cost. Thus $C_F(0) = 0$. Begin with $F = 9$ and $a = 12$. In the first model the active mass n of identical atomistic plants may be any nonnegative real number. A free entry competitive equilibrium requires individual global profit maximization, market clearing, and zero maximal profit when $n > 0$, since further identical plants may enter. Ignore financing constraints and externalities.

(a) For every $p \geq 0$, find a plant's globally optimal supply correspondence. Explain why solving a first order condition conditional on opening does not determine whether the plant should open. Include the price at which the plant is indifferent between opening and staying closed. [10]

(b) For $F = 9$ and $a = 12$, find the equilibrium price, output per active plant, active mass, and total output. Verify all the equilibrium conditions. [8]

(c) A production improvement reduces F from 9 to 1 without changing variable cost. Find the new equilibrium. Calculate the change in consumer surplus and total surplus, counting every active plant's setup cost. Explain how output per plant, total output, and the active mass can move differently. [10]

(d) Now change the model. There are countably many potential plants, the number n of active plants must be an integer, and every active plant remains a price taker. Retain the equilibrium requirement that no inactive plant can earn strictly positive profit at the prevailing price. Set $F = 9$ and $a = 13$. Does such a competitive free entry equilibrium exist? Prove your answer. Explain which divisibility assumption in the first model matters and why an integer entry game would be a different model. [12]

Optional tool extension. For a grid of a and F, compute the divisible mass equilibrium and then test whether it can be implemented by an integer number of identical plants under the exact prevailing price definition in part d. Keep this model separate from a strategic entry game in which an entrant anticipates a new price. Use exact arithmetic when a and $\sqrt{F}$ are rational.

Problem 10.A4 [Advanced]. Two local markets, a capacity bottleneck, and investment.

Suggested time: 45 minutes. Total: 40 points.

Two locations A and B consume a homogeneous service. Their inverse demands are $P_A(q_A) = 10 - q_A$ and $P_B(q_B) = 14 - 2q_B$ over nonnegative quantities, with zero purchases at prices at or above their respective choke prices. They arise from utilities $c_A + 10q_A - q_A^2/2$ and $c_B + 14q_B - q_B^2$, with money sufficiently abundant that its nonnegativity never binds. All production occurs at A. A price taking producer chooses total output $Q \geq 0$ and pays $C(Q) = 2Q + Q^2/2$. An existing transporter buys t units at price p_A and resells them at p_B, at no variable resource cost, subject to $0 \leq t \leq k$, where $k > 0$. It takes both prices as given. Clearing requires $Q = q_A + t$ and $t = q_B$. The producer and transporter maximize profits. There are no taxes or externalities.

(a) Write the surplus maximization problem over $q_A, q_B \geq 0$ with $q_B \leq k$. Derive a KKT certificate, prove global optimality, and find the allocation as a function of $k > 0$. Identify the exact capacity threshold beyond which additional capacity is unused. [12]

(b) Find prices and profits that decentralize your allocation as a competitive equilibrium. Interpret $p_B - p_A$ and verify the transporter's choice in both the binding and slack capacity regimes. Give all quantities, prices, and the transporter's profit at $k = 1$. [10]

(c) Compare $k = 1$ with $k = 3$. Calculate the change in total surplus and in each location's consumer surplus. Does greater connectivity necessarily make both locations' consumers better off? Explain how this can coexist with a total surplus gain. [8]

(d) A planner can expand the original $k = 1$ link by $e \geq 0$ at real cost e^2, so $k = 1 + e$. Find the globally optimal expansion and verify the regime used. Relate the expansion condition to the capacity price gap. Explain why the transporter's lost or gained rent is not an additional resource cost of the project. [10]

Optional tool extension. Extend the planner model to three locations and two directed links. List flow conservation and link capacity constraints explicitly. Enumerate active links, derive nodal prices from marginal benefits and production costs, and check every zero flow or capacity inequality. First reproduce the two location solution exactly. A solver's allocation is not sufficient without those checks.

Problem 10.A5 [Advanced]. The original continuum market: population, productive capacity and welfare.

Restored from the Western University Fall 2019 final examination. The original model and five part structure are retained. The two interpretations of producer selection are stated separately, and the welfare comparison holds initial wealth fixed.

Suggested time: 60 minutes. Total: 60 points. For a timed rehearsal, attempt parts (a) to (d) in 45 minutes and complete part (e) afterward.

Consider a competitive market for a consumption good x and a numeraire m. There is a mass $\eta \geq 1$ of consumers. A consumer has taste $\alpha \sim U[0, 1]$, initial numeraire wealth $\omega \geq 0$, and utility $u_\alpha(x, m) = 2\sqrt{\alpha x} + m$. The joint distribution F of (α, ω) is held fixed across comparisons, and its mean wealth is $\Omega < \infty$. The consumer chooses $x \geq 0$ and unrestricted $m \in \mathbb{R}$ subject to $px + m \leq \omega$. Thus borrowing is allowed, the initial endowment of the consumption good is zero, and initial wealth is a fixed partial equilibrium primitive. Do not add producer profit to this consumer budget.

Initially there is a unit mass of competitive firms. Their productivity e is uniformly distributed on $(0, 1)$, and a type e firm chooses output $q \geq 0$ with numeraire cost $C(q; e) = q^2/(2e)$. The output price is $p > 0$ and the numeraire price is 1. There is no fixed cost or entry margin.

(a) Find each consumer's goods demand and each firm's supply. Aggregate over both populations and solve for the equilibrium price and total quantity. Distinguish a mean quantity from a population total, and justify individual optimality and equilibrium uniqueness. [14]

(b) Derive the individual profit function $\pi(p, e)$ and verify Hotelling's lemma by differentiating with respect to the output price. [8]

(c) Consider the original question's earthquake interpretation: all firms with $e < 1/2$ disappear without replacement. State the mass of surviving firms, derive aggregate supply, and find the new equilibrium price and quantity. Now consider a different experiment: replace the initial firms by a unit mass whose productivity is uniformly distributed on $[1/2, 1]$. Solve this experiment as well. Compare both prices with the initial equilibrium and explain why a higher conditional mean productivity does not by itself determine the effect of an earthquake. [16]

(d) At the initial equilibrium, derive the cross sectional variance of an individual consumer's goods consumption. How does it change with η when the producer population and the joint type and wealth distribution are held fixed? This part uses the absolute variance, not a normalized relative inequality measure. [10]

(e) Derive average indirect utility and average remaining numeraire consumption at the initial equilibrium. How do they change with η? Reconcile the welfare comparison with part (d), and explain why holding initial wealth fixed is different from holding remaining numeraire consumption fixed. [12]

Problem 10.A6 [Advanced]. Heterogeneous elasticities, a hidden aggregation identity and ownership income.

Restored from the first exercise of the SFU Fall 2024 Problem Set 7, also used at Western in 2020. The continuum model and ownership accounting are retained. The endpoint utility is normalized, and the price limit and resource accounting are corrected. Part (d) and the optional tool extension are new practice extensions.

Suggested time: 60 minutes. Total: 60 points. Work by hand.

There is a unit mass of consumers indexed by (α, ω). Their joint probability distribution is F, its taste marginal is $\alpha \sim U[1, 2]$, and the initial numeraire endowment $\omega \geq 0$ has finite mean Ω. A consumer chooses a consumption good $x > 0$ and numeraire $m \in \mathbb{R}$. The numeraire can be negative, so no borrowing or nonnegative money constraint is imposed. For a fixed demand parameter $A \in (0, 1]$, utility is $u_\alpha(x, m) = \phi_\alpha(x; A) + m$, where $\phi_\alpha(x; A) = A^{1/\alpha} \frac{\alpha}{\alpha-1}(x^{1-1/\alpha} - 1)$ for $\alpha > 1$ and $\phi_1(x; A) = A \log x$.

There is also a unit mass of competitive firms with type $\beta \sim U[1, 2]$. A type β firm chooses $q \geq 0$ and uses the numeraire input $C_\beta(q) = q^{1+1/\beta}/(1 + 1/\beta)$. The consumption good price is $p > 0$ and the numeraire price is 1. There are no fixed costs or entry choices. Every firm is owned by consumers. The ownership kernel is a measurable function $\theta(\alpha, \omega, \beta) \geq 0$ satisfying $\int \theta(\alpha, \omega, \beta) dF(\alpha, \omega) = 1$ for every β. The consumer's income is $y(\alpha, \omega; p) = \omega + \int_1^2 \theta(\alpha, \omega, \beta)\pi_\beta(p) d\beta$, and the budget is $px + m \leq y$. Assume the displayed integrals are finite. The initial endowment of the consumption good is zero.

(a) Verify the utility limit at $\alpha = 1$. Derive the consumer's unique demand and the firm's unique output and profit at every $p > 0$. Give global optimality arguments, not only first order equations. Explain why ownership income does not affect the goods demand under the stated budget domain. [10]

(b) Define $H(p) = \int_1^2 p^t dt$. Derive its expression for $p \neq 1$, its value at $p = 1$, and $H'(1)$. Express aggregate goods demand $D(p)$ and supply $S(p)$ using H. Find an identity relating $D(p)$, $S(p)$ and p that remains valid at $p = 1$. Explain which feature of the two type distributions makes this identity possible. [14]

(c) Find the unique equilibrium price and total quantity for every $A \in (0, 1]$. Verify the case $A = 1$ directly instead of canceling terms containing $\log p$. Derive each type's equilibrium quantity and show how consumer and producer types can be paired to check market clearing. [10]

(d) Compute the price elasticity with respect to A. Express the elasticity of total equilibrium output with respect to A as a weighted average of producer types, and derive meaningful bounds. Evaluate its limiting value as A approaches 1. Explain why individual constant elasticities do not imply a constant aggregate quantity elasticity along this equilibrium path. Do not infer a welfare ranking merely from changing a taste parameter. [12]

(e) Derive mean numeraire consumption, including distributed profits. Prove that average consumption of the numeraire plus the numeraire used in production equals its initial aggregate endowment. At $A = 1$, evaluate total profit, total production cost and mean numeraire consumption. Explain precisely which assumption supports the claim that the distribution of initial wealth and ownership does not change the equilibrium goods price, and why that claim can fail if consumers must have $m \geq 0$. [14]

Optional tool extension. First replace the uniform laws by a consumer elasticity distribution G on $[1, 2]$ and a producer distribution equal to the law of $3 - \alpha$ when $\alpha \sim G$. Derive the new aggregate identity and price for consumer mass $M_c > 0$ and firm mass $M_f > 0$. State how the ownership normalization must change when the consumer measure has mass M_c. Then choose a producer distribution that is not this reflection and test whether the same closed form still holds. Verify individual budgets, ownership totals, both aggregate resource equations and the $p = 1$ limits. Numerical integration is a check, not a substitute for the aggregation proof.

Problem 10.A7 [Advanced]. How market clearing creates endogeneity.

Restored from Exercise 2 of the SFU Fall 2024 Problem Set 7, also used at Western in 2019. The original static market model and four identification tasks are retained. Slope terminology, stochastic assumptions and the excluded supply shifter are corrected. Part (e) and Part II are new time series extensions.

Suggested time: 50 minutes for Part I. Total: 60 points. Work by hand before using a program.

Part I. By hand

At each date t, a competitive market clears after a demand shock u_t and a supply shock v_t are realized. Good 2 is the numeraire; p_t is the price of good 1 in units of good 2, and Q_t is its traded quantity. Aggregate demand and supply on their interior ranges are $Q_t^d = \alpha_0 + \alpha_1 p_t + u_t$ and $Q_t^s = \beta_0 + \beta_1 p_t + v_t$, where $\alpha_1 < 0 < \beta_1$ and all four coefficients are fixed. There is no storage or choice linking dates. Equilibrium requires $Q_t = Q_t^d = Q_t^s$. Assume a bounded shock support on which the clearing price and quantity are strictly positive; state the required inequalities in part (a). Do not truncate negative computed observations or replace them by zero.

Initially, (u_t, v_t) are independent and identically distributed across dates, have mean zero, satisfy $\mathrm{Cov}(u_t, v_t) = 0$, and have finite variances σ_u^2, σ_v^2 with $\sigma_u^2 + \sigma_v^2 > 0$. The shocks need not

be independent within a date. Write $D = \beta_1 - \alpha_1 > 0$. An analyst observes only equilibrium (p_t, Q_t) and regresses Q_t on a constant and p_t. Let $\widehat{\gamma}_{1,T}$ denote the ordinary least squares slope from T observations, using sample covariances with sample means removed.

(a) Derive the unique clearing price and quantity for every shock realization. State the two inequalities defining the strictly positive equilibrium region and verify both original market equations. Explain why observations from different dates do not generally trace one fixed demand curve. [12]

(b) Derive $\mathrm{Cov}(p_t, u_t)$, $\mathrm{Cov}(p_t, v_t)$, $\mathrm{Var}(p_t)$ and the probability limit of $\widehat{\gamma}_{1,T}$. Express that limit as a weighted average of the demand and supply slopes. Calculate its difference from α_1. Explain the economic source of the correlation rather than calling it only an omitted variable problem. Distinguish a slope from the price elasticity of demand at an interior observation. [16]

(c) Characterize exactly when this regression identifies the demand slope under the maintained assumptions, and when it instead identifies the supply slope. What if neither shock varies? For $\alpha_1 = -2$, $\beta_1 = 1$, $\sigma_u^2 = 4$ and $\sigma_v^2 = 1$, determine the sign and value of the limiting regression slope. Can the fitted relationship slope upward even though every demand curve slopes downward? [8]

(d) Replace the supply shock by $v_t = -\delta z_t + \eta_t$, where $\delta > 0$ and z_t is an observed cost shifter. For this part, u_t, η_t, z_t are mutually independent, centered and independent across dates, with finite variances and $\mathrm{Var}(z_t) > 0$. Retain the positive equilibrium support restriction. Derive the covariance of z_t with equilibrium price and quantity. Show which slope is identified by their ratio and justify consistency of the corresponding sample covariance estimator. Give a plausible economic cost shifter and state its exclusion restriction. Explain why a shifter that directly enters demand is not generally an excluded instrument for demand. Which slope would a pure demand shifter identify under the symmetric exclusion assumptions? [12]

(e) New time series extension. Keep the static market and the cost shifter formulation, but let $u_t = \rho_u u_{t-1} + \varepsilon_t$, $\eta_t = \rho_\eta \eta_{t-1} + \xi_t$ and $z_t = \rho_z z_{t-1} + \omega_t$, where $|\rho_u|, |\rho_\eta|, |\rho_z| < 1$. Each process is initialized in its stationary law. The three innovation sequences are mutually independent across processes and dates, centered, bounded and have finite variance. The three stationary processes are independent of one another; their supports and intercepts ensure positive clearing prices and quantities. You may use the stationary moment law of large numbers for these processes. Does contemporaneous z_t remain a valid demand instrument? Derive $\mathrm{Cov}(p_{t-1}, u_t)$ and assess whether lagged price is automatically a valid and relevant instrument. Explain why correcting standard errors for serial correlation cannot repair an invalid instrument. Does the simultaneity problem here require stochastic trends or dynamic optimization? [12]

Part II. Codex and verification

Seed: the clearing equations and population moments from Part I. Fixed: the static economic model, coefficients, marginal shock variances, independence across the three shock processes, sample length and estimator definitions. Scaled: a common persistence parameter $\rho_u = \rho_\eta = \rho_z = \rho$ over $0, 1/4, 1/2, 3/4, 4/5$. Risk: confusing serial dependence with identification, using a weak covariance denominator, or silently changing the economic model by clipping observations.

Use $\alpha_0 = 100$, $\beta_0 = 0$, $\alpha_1 = -2$, $\beta_1 = 1$, $\delta = 1$, and stationary marginal variances $\mathrm{Var}(u_t) = \mathrm{Var}(\eta_t) = \mathrm{Var}(z_t) = 1$. One bounded implementation uses independent innovations taking the two values $\pm\sqrt{1-\rho^2}$ with equal probability for each process. Document stationary initialization or a controlled burn period. Keep $T = 20000$ retained observations per run and use

several recorded random seeds.

Compute clearing prices and quantities directly, verify both market equations and prove that the prescribed support keeps them positive. Compare ordinary least squares, the current cost instrument and the lagged price instrument with the population limits derived by hand. At $\rho = 0$, report a population first stage that vanishes as irrelevance, not as a numerical estimate of a structural slope. Report covariance denominators and sampling variation; if reporting standard errors, use a procedure valid for dependent moment observations.

Include a deliberately invalid instrument $z_t^{bad} = z_t + u_t$ and explain which moment restriction fails. Distinguish the finite stationary experiment from a spurious regression between unrelated stochastic trends. Your program must reproduce the independent date benchmark and the analytical moment identities before presenting graphs. A simulation does not establish an exclusion restriction or a continuum of parameter claims.

Problem 10.A8 [Advanced]. Prices, quality and a small differentiated product market.

This original teaching case connects stochastic choice to firms that set prices. It is inspired by Berry, Levinsohn and Pakes, Automobile Prices in Market Equilibrium, Econometrica 1995, DOI 10.2307/2171802. The hand calculation is a homogeneous logit benchmark, not the full BLP model or its automobile application.

Part I. By hand

Allow 50 minutes. There are 50 practice points. In each market t, a unit mass of consumers chooses at most one of two inside products $j = 1, 2$, or the outside option $j = 0$. Product j has an observed scalar characteristic x_{jt}, coefficient β, and quality ξ_{jt} that consumers and firms know but an analyst may not observe. Write $\delta_{jt} = \beta x_{jt} + \xi_{jt}$. Consumer i obtains $u_{ijt} = \delta_{jt} - \alpha p_{jt} + \varepsilon_{ijt}$ from an inside product and $u_{i0t} = \varepsilon_{i0t}$ from the outside option. Here $p_{jt} \geq 0$ is a price, $\alpha > 0$ is the marginal utility cost of a unit of expenditure, and all ε_{ijt} are independent across consumers, products and markets with distribution function $F(e) = \exp\{-\exp(-e)\}$. The outside option has price and deterministic utility zero. Consumers take prices as given; income constraints do not bind in this quasilinear demand specification.

You may use the logit formula: if $v_{jt} := \delta_{jt} - \alpha p_{jt}$, then $s_{jt} = e^{v_{jt}}/(1 + e^{v_{1t}} + e^{v_{2t}})$ and $s_{0t} = 1/(1+e^{v_{1t}}+e^{v_{2t}})$. These shares are probabilities over the whole market, including consumers who buy neither inside product.

(a) Derive $\log(s_{jt}/s_{0t})$ and all own and cross price derivatives of the shares, holding qualities fixed. For $k \neq j$, including $k = 0$, define the diversion ratio from j to k as $D_{j\to k} := (\partial s_{kt}/\partial p_{jt})/(-\partial s_{jt}/\partial p_{jt})$. Derive it and explain what the formula does, and does not, say about products that are near one another in characteristic space. [12 points]

(b) Each product is sold by a different firm. Firm j has constant marginal cost $c_j > 0$, no fixed cost or capacity constraint, and chooses $p_j \geq 0$ to maximize $\pi_j = (p_j - c_j)s_j$. It observes qualities and takes the rival price as given. Derive its interior pricing condition and express the markup $p_j - c_j$ using α and its share. Then prove that the condition identifies the unique global best response, not merely a stationary price. You may first exclude prices below cost and then study how the sign of the profit derivative changes. Do not assume that the profit function is globally concave. [14 points]

(c) Set $\alpha = 1$, $\beta x_{jt} = 0$ and $c_1 = c_2 = 1/2$ in both markets. In market 1, both qualities are $\xi_{j1} = 2$ and a proposed price profile is $p_{11} = p_{21} = 2$. In market 2, both qualities are $\xi_{j2} = 9/4 + \log 3$ and a proposed profile is $p_{12} = p_{22} = 9/4$. Compute all three shares, each firm's markup and profit in each market. Verify that both profiles are Nash equilibria using the global best response argument. Within each market, hold its supplied qualities fixed when considering a price deviation. [12 points]

(d) An analyst pools the four product observations and runs an ordinary least squares regression with an intercept of $y_{jt} := \log(s_{jt}/s_{0t})$ on p_{jt}, omitting quality. Calculate the fitted price slope exactly. Compare it with the structural price coefficient in utility. Explain how a higher price and a higher share can occur together across these markets without violating downward sloping demand. State an economically plausible excluded cost shifter, the exclusion and relevance conditions it would need, and why two constructed markets do not establish those conditions. [12 points]

Part II. Codex and verification

Scale one feature: replace the common price coefficient by two known consumer types, with $\alpha_1 = 1/2$ and $\alpha_2 = 3/2$, each of mass $1/2$. Hold the two products, two markets, recorded prices, qualities, outside option and shock distribution fixed. This is a conditional demand inversion exercise. Do not assume that the old prices remain pricing equilibria after the preference distribution changes.

Let $\bar{\alpha} = 1$ and define the mean utility net of the mean price effect by $v_{jt} = \delta_{jt} - \bar{\alpha} p_{jt}$. Type h has deterministic inside utility $v_{jt} - (\alpha_h - \bar{\alpha}) p_{jt}$. Compute its logit probabilities, including the outside option, and average over types to obtain aggregate shares $s_{jt}(v_t, p_t)$. First generate exact model target shares S_{jt} from the supplied qualities and prices. Keep the known type distribution fixed throughout inversion.

Write a small program at `work/blp_case.py` that recovers v_t from the target inside shares and the recorded prices. You may use the iteration $v_{jt}^{r+1} = v_{jt}^{r} + \log S_{jt} - \log s_{jt}(v_t^r, p_t)$, where r is an iteration count. State initialization, stopping rule, tolerance and maximum iteration count. Compute probabilities stably and keep the outside deterministic utility fixed at zero. Report a failure rather than a recovered answer if the stopping conditions are not met.

Run `python3 work/blp_case.py --check`. The check must compare recovered mean utilities with the values used to generate the targets, reconstruct all inside and outside shares, verify probability adding up, and test several distinct initial vectors. Require maximum share error below 10^{-10} and maximum mean utility error below 10^{-8}. Include the homogeneous limit in which both price coefficients equal one, one deliberately invalid target with zero outside share that the input checks reject, and a demonstration that the homogeneous log odds formula need not invert the mixed model. Numerical success is not a proof of a global contraction on all utility vectors.

Before coding, record the hand seed, fixed assumptions, the single heterogeneity extension and its new failure risk. Submit source, material prompts, the validation command and output, and one generated step checked independently. Finally explain why share inversion conditional on known taste parameters does not itself identify unknown taste parameters or solve price endogeneity. Discuss what additional markets, excluded cost variation and rank conditions an actual estimation design would require. Estimating the full BLP model, simulating an automobile

industry and proving a general identification theorem are not part of this exercise.

Part II: Formal and applied extensions

Problem 10.2 [Applied]. A per-unit tax.

Introduce a per-unit tax $t > 0$ paid by sellers into the market of Problem 10.1, so buyers pay p_b and sellers receive $p_s = p_b - t$. Assume t is small enough that equilibrium quantity remains nonnegative.

(a) Solve for p_b, p_s, and quantity.

(b) Derive tax incidence. Express the local burden shares using the demand and supply slopes, and interpret them using point elasticities evaluated at the common no-tax equilibrium.

(c) Compute deadweight loss and its derivative with respect to t.

Problem 10.3 [Proof]. Planner and competitive allocation.

A representative consumer has quasilinear utility $u(q)+m$, with enough transferable numeraire that the optimum is not constrained by a lower bound on m. The functions u and C are differentiable on $(0, \infty)$, have the required right derivatives at zero, u is increasing and strictly concave, and C is increasing and convex. Assume the surplus maximum over $q \geq 0$ is attained.

(a) State and solve the planner's problem using Kuhn-Tucker conditions.

(b) Show that any interior competitive equilibrium satisfies the same condition.

(c) State the assumptions required for the converse decentralization claim.

Music sales and concerts

Chapter use case. Seed: Problems 1.2, 9.1, 10.1, 10.3. Fixed: all unlisted primitives and assumptions. Scaled: the music-volume or royalty share τ from the two benchmark values to a grid. Tool: Codex + Python. Verification: Reproduce the demand and cost formulas and both endpoint comparisons; bracket every stationary point, check both price boundaries, establish the global maximizer, and keep the volume-share and royalty-share scenarios distinct.

Problem 10.4 [Applied]. The Economics of Superstars: Music Sales and Concerts.

Taylor S. is a stylized superstar whose fans consume music services and concerts. This is the graduate version of the music-and-concerts case, with explicit preferences, technology, and market assumptions. All numbers are hypothetical; the model is not an empirical description of Taylor Swift's contracts. Quantities are divisible, and the mass of identical fans is one.

Part I. Fan demand

Write q_M, q_C for music and concert services, the two coordinates of consumption. A fan has wealth $w > 0$, prices $p_M, p_C > 0$, and utility $u(q_M, q_C) = (\frac{1}{2}q_M^\rho + \frac{1}{2}q_C^\rho)^{1/\rho}$, where $0 < \rho < 1$. The letters M, C are local commodity labels.

(a) Formulate the budget problem and derive demands using $\sigma = 1/(1-\rho)$. For the remaining parts set $\rho = 1/2$, $w = 1$, and $p_M = 1$. Explain why music demand changes when the concert price changes.

Part II. Production

For $j \in \{M, C\}$ the artist's technology is $y_j = A_j\sqrt{n_j}$, where $n_j \geq 0$ is labor and $A_j > 0$ productivity. Labor costs $\nu > 0$ per hour, distinct from fan wealth w. The processes are independent. This stylized music-service cost does not describe the cost of creating an original recording as a function of its streams.

(b) Derive $C_j(\nu, y_j)$, average cost, and marginal cost. Explain decreasing returns. Set $A_M = A_C = \nu = 1$ in the numerical parts.

Part III. A share of volume and receipts

An exogenous $\tau \in (0, 1]$ is the fraction of music consumption attributable to this artist. Her output is $y_M = \tau q_M$ and her music receipts are $\tau p_M q_M$. Concert output is $y_C = q_C$. She takes p_M, τ as given and chooses $p_C > 0$. There is no shared labor constraint or fixed cost.

(c) Write total revenue and profit after substituting demand and costs. Distinguish a revenue derivative with respect to ticket price from marginal revenue with respect to output. Account for changes in both revenue streams and production costs.

(d) Derive the first-order condition for the optimal concert price. Examine profit as $p_C \downarrow 0$ and $p_C \to \infty$. Establish global optimality and uniqueness, analytically or with a justified numerical method accounting for every stationary point and boundary.

(e) Compare $\tau = 1$ and $\tau = 1/2$. In a table report optimal concert prices, both fan quantities, both delivered outputs, revenues, costs, and profits. State numerical accuracy. Does the concert price rise or fall when the music share falls?

(f) Use the first-order condition or implicit differentiation to analyze the price response for $\tau \in [1/2, 1]$. Interpret the roles of substitution and production costs. Do not assume the same sign for every technology or parameter value.

Part IV. A royalty share

Now τ changes only the fraction of music revenue paid to the artist: music output is $y_M = q_M$, while receipts remain $\tau p_M q_M$. All other assumptions are unchanged.

(g) Write the revised profit function and repeat the comparison between $\tau = 1$ and $\tau = 1/2$. Identify the changed cost channel and determine whether the earlier price response survives.

(h) Explain what evidence distinguishes a fall in music volume from a fall in the royalty fraction. Discuss the roles of CES substitution, decreasing returns, and control of concert prices. Contrast the artist's decision with price-taking supply in the main chapter.

Chapter 11

General Equilibrium

11.1 Exchange, production and welfare

Part I: By hand

Problem 11.1 [Core]. Cobb-Douglas exchange equilibrium.

There are two consumers with common utility $u_i(x_{i1}, x_{i2}) = x_{i1}^{\alpha} x_{i2}^{1-\alpha}$, $0 < \alpha < 1$, and endowments $\omega_1 = (1, 2)$, $\omega_2 = (2, 1)$.

(a) Derive individual Walrasian demands.

(b) Normalize $p_2 = 1$ and solve for the equilibrium relative price.

(c) Compute equilibrium allocations and verify both market-clearing equations.

Advanced exam practice

The suggested time applies to the hand questions, not the optional tool extension. Start with a diagram or solution plan, state the constraints, check all regimes and explain global optimality and economic accounting. Attempt the hand questions before using tools. These problems are additional preparation, not extra weekly requirements unless assigned in class. Before an optional extension, state Seed, Fixed, Scaled and Risk and enlarge one feature at a time.

Difficulty guide. Problems 11.A1 to 11.A6 develop geometric and regime reasoning. Problem 11.A7 restores the Western 2018 final examination question on the empirical content of equilibrium with production. It requires a full mathematical certificate and both directions of a rationalization argument, not only a computed equilibrium.

Problem 11.A1 [Advanced]. The lens, the strong core and the price line.

Suggested time: 40 minutes for the hand part. Total: 50 points. Work by hand.

There are two consumers A and B, two goods, consumption sets $\mathbb{R}^2_+$, and utilities $u^i(x) = \sqrt{x_1 x_2}$. Endowments are $\omega^A = (5, 1)$ and $\omega^B = (1, 5)$. Prices are strictly positive. For this question, a coalition strongly blocks an allocation if it can redistribute its own endowment so that every member is weakly better off and at least one is strictly better off; the strong core admits no such coalition.

(a) Draw the box, mark both origins and the endowment point, and express B's consumption when A receives (a, b). State the resource equations and the inequalities defining the individually rational lens. [8]

(b) Derive the complete Pareto set. Give an explicit mutually beneficial feasible trade from every allocation with $a \neq b$, rather than relying only on tangency. [10]

(c) Determine the strong core exactly. Explain why its characterization uses both singleton coalitions and the coalition containing both consumers. [8]

(d) Normalize $p_2 = 1$. Derive demands, solve for the equilibrium relative price and allocation, and check both budgets and both markets. Mark the common budget line in your diagram. [10]

(e) Can every allocation in the strong core be a competitive allocation with the original endowments? Justify your answer. Find transfers with $T_A + T_B = 0$ that support $x^A = (7/2, 7/2)$ and the complementary bundle for B. [8]

(f) Explain which of the box, Pareto set, individually rational lens and competitive allocation can change when physical endowments are redistributed while their aggregate is held fixed. [6]

Optional tool extension. Keep utilities and aggregate resources fixed and vary only the scalar $s \in [0, 6]$ in endowments $(s, 6 - s)$ and $(6 - s, s)$. Use Codex and Python to draw the lens and strong core and recover the equilibrium for a rational grid including $s = 0, 3, 6$. Verify both budgets and resources exactly, and verify individual rationality through squared utilities. Distinguish these finite checks from the hand proof.

Problem 11.A2 [Advanced]. A kink, an indifferent consumer and one clearing allocation.

Suggested time: 40 minutes for the hand part. Total: 50 points. Work by hand.

Consumption sets are $\mathbb{R}^2_+$. Consumer A has $u^A(x) = \min\{x_1, x_2\}$ and endowment $(4, 0)$. Consumer B has $u^B(x) = 2x_1 + x_2$ and endowment $(0, 4)$. Normalize $p = (q, 1)$ with $q > 0$. A competitive equilibrium requires individual utility maximization and exact clearing in both goods.

(a) Derive A's demand and B's full demand correspondence for every $q > 0$. Explain why applying a differentiable tangency condition to A or selecting a single arbitrary demand for B can give a wrong answer. [12]

(b) Show that no equilibrium can have $q < 2$ or $q > 2$. At the remaining relative price, select demands that clear both markets. Verify budgets and identify whether prices, equilibrium allocations and individual demand are unique. [12]

(c) Draw the box, both origins, the endowment, the equilibrium and the common budget line. At $q = 1$ and $q = 3$, compute aggregate excess demand and explain the direction in which relative prices would have to move to reach equilibrium. [8]

(d) Derive the entire Pareto set using a direct argument. Which resources can A give away without lowering utility when $x^A_1 \neq x^A_2$? [10]

(e) Under the book's strong blocking convention, determine the strong core. Compare it with the competitive allocation and explain what this example teaches about kinks, corner choices and market clearing. [8]

Optional tool extension. Vary only q on a positive rational grid containing $1, 2, 3$ and values immediately on both sides of 2. Compute the full demands and check $qz_1 + z_2 = 0$ exactly. At $q = 2$, represent B's full demand segment and recover the clearing selection rather than inserting an arbitrary bundle. A plot of excess demand is not a proof of price adjustment convergence.

Problem 11.A3 [Advanced]. An endowment threshold makes a consumer enter.

Suggested time: 45 minutes for the hand part. Total: 50 points. Work by hand.

There are two goods, denoted (x, y), and two consumers with nonnegative consumption. Preferences are $u^A(x, y) = \log(1 + x) + y$ and $u^B(x, y) = 2\log(1 + x) + y$. Endowments are $\omega^A = (Q, 0)$ and $\omega^B = (0, 10)$, where $Q > 0$. Normalize prices to $(q, 1)$ with $q > 0$. Consumers take prices as given.

(a) Derive both demands for every $q > 0$, including A's possible zero consumption of y and zero consumption of x. Do not assume the numeraire is positive before checking it. [12]

(b) Find all competitive equilibria for every $Q > 0$. Give the exact threshold at which A starts consuming good x, and report all four quantities and the relative price on both sides of the threshold, including equality. [14]

(c) Verify both budgets and both markets. Give a global optimality argument for each consumer and evaluate the entry condition for A's good x at the threshold and on either side. [10]

(d) Compute the equilibrium for $Q = 1/2$ and $Q = 4$. Draw the corresponding boxes, with good x horizontal. Mark the equilibrium corner in the first box and the interior allocation in the second. [8]

(e) Is the equilibrium relative price continuous and differentiable at the entry threshold? Explain why a zero choice at an equality threshold is compatible with a unique optimum, and why quasilinearity alone does not justify ignoring the nonnegative numeraire constraint. [6]

Optional tool extension. Vary only $Q > 0$ on a rational grid containing $1/2, 1, 4$ and values immediately around 1. Implement the full capped demands and the equilibrium formulas independently. Check nonnegativity, both budgets, both market equations, entry inequalities and complementary products exactly. Plot the entry threshold and explain the different one sided derivatives.

Problem 11.A4 [Advanced]. Redistribution moves prices but does not resize the box.

Suggested time: 40 minutes for the hand part. Total: 50 points. Work by hand.

Two consumers have $u^A(x) = x_1^{2/3}x_2^{1/3}$ and $u^B(x) = x_1^{1/3}x_2^{2/3}$ on $\mathbb{R}^2_+$. Endowments are $\omega^A = (t, 3)$ and $\omega^B = (6-t, 3)$, where $0 \le t \le 6$. Normalize prices to $(q, 1)$ with $q > 0$. Increasing t transfers physical units of good 1 from B to A before trade.

(a) Derive both demands as functions of q and t. Solve for the unique equilibrium relative price and all four consumption quantities. Check both resource equations. [16]

(b) Compute the equilibria at $t = 3$ and $t = 6$. Explain why a transfer of a good can raise that good's relative price even though aggregate resources have not changed. [8]

(c) Using A's coordinates (a, b), derive the interior contract curve and include its endpoints with a valid boundary argument. Draw the two equilibria on the same box. [10]

(d) A policymaker proposes the equal allocation $x^A = x^B = (3, 3)$. Can any lump sum transfers and common strictly positive prices decentralize it as an equilibrium for these preferences? Give an explicit Pareto comparison rather than relying only on unequal marginal rates of substitution. [10]

(e) Distinguish a redistribution of initial physical endowments from a change in the aggregate endowment. Explain which objects remain fixed here and why calculating a price effect while holding both consumers' wealths fixed would answer a different question. [6]

Optional tool extension. Vary only $t \in [0, 6]$ on a rational grid containing $0, 3, 6$. Verify the two budget identities, both resource equations and membership of the reported allocation in the

contract curve exactly. Plot $q(t)$ and the equilibrium path on a single unchanged box. Do not introduce a second taste or resource parameter.

Problem 11.A5 [Advanced]. Who owns the firm and who receives its gains.

Suggested time: 45 minutes for the hand part. Total: 50 points. Work by hand.

There are two goods and two consumers, each endowed with $(2, 0)$ and each having utility $u^i(x) = \sqrt{x_1 x_2}$ on $\mathbb{R}^2_+$. A firm can use $z \geq 0$ units of good 1 to produce $2\sqrt{z}$ units of good 2. A owns share $\theta \in [0, 1]$ of its profit and B owns share $1 - \theta$. Normalize prices to $(q, 1)$ with $q > 0$. The firm and consumers take prices as given.

(a) Solve the firm's profit maximization problem for every $q > 0$, prove global optimality and report input, output and profit. Explain why diminishing returns can generate positive profit without monopoly power. [8]

(b) Write both consumers' wealths and derive their demands. Find the equilibrium price, production and consumption for every $\theta \in [0, 1]$. [14]

(c) Verify both consumer budgets and the two resource equations, including the input used by the firm. Explain why good 1 consumption does not sum to its initial endowment. [8]

(d) Compare $\theta = 0$ and $\theta = 1$. In this model, which real decisions depend on ownership and which do not? Identify the preference feature responsible for the aggregate invariance, without claiming that ownership never matters for equilibrium production. [6]

(e) A planner uses a fixed weight $\gamma \in (0, 1)$ and maximizes $\gamma(\log x_1^A + \log x_2^A) + (1-\gamma)(\log x_1^B + \log x_2^B)$ subject to the production and resource constraints. Solve first for the allocation conditional on z, then for z. Find the weight supporting the competitive allocation for a given θ. [10]

(f) After production is fixed, draw the feasible exchange box for the output and remaining input. Explain why using the original 4 by 0 endowment rectangle as the final consumption box would be misleading. [4]

Optional tool extension. Vary only $\theta \in [0, 1]$ on a rational grid containing $0, 1/2, 1$. Check both ownership shares, both private budgets and both resource equations, retaining $\sqrt{3}$ symbolically or stating numerical tolerances. Verify the corresponding planner weight and distinguish distributional changes from the invariant production and price choices.

Problem 11.A6 [Advanced]. A tax wedge remains even when every dollar is rebated.

Suggested time: 45 minutes for the hand part. Total: 50 points. Work by hand.

Two consumers have utility $u^i(x, y) = \sqrt{xy}$ on $\mathbb{R}^2_+$. Endowments are $\omega^A = (4, 0)$ and $\omega^B = (0, 4)$. The producer prices are $(q, 1)$ with $q > 0$. Consumer A pays an ad valorem tax $\tau \geq 0$ on every unit of good y consumed. Consumer B is not taxed and receives all tax revenue as a lump sum transfer T. The government consumes no goods. Each consumer takes (q, T) as given. Their budgets are $qx_A + (1 + \tau)y_A \leq 4q$ and $qx_B + y_B \leq 4 + T$. Government budget balance requires $T = \tau y_A$.

(a) Derive both demands conditional on (q, T). State the exact equations for clearing the goods markets and balancing the government budget. [8]

(b) Solve for q, T and all four consumption quantities as functions of τ. Verify both goods markets and both private budgets. Do not count tax revenue as destroyed physical output. [14]

(c) Draw the equilibria at $\tau = 0$ and $\tau = 2$ on the same Edgeworth box. Explain why A and B generally do not face the same budget slope even though there is one producer price vector. [8]

(d) Compare both consumers' welfare as τ rises from zero. Explain why a policy can benefit its rebate recipient but still generate an inefficient allocation. [6]

(e) For every $\tau > 0$, construct a feasible allocation that makes both consumers strictly better off than the taxed equilibrium. Verify the gains algebraically, including nonnegative consumption. You may look for a trade that moves each consumer toward equal quantities of the two goods. [10]

(f) Which hypothesis of the competitive welfare argument fails? Distinguish a transfer of purchasing power from the marginal tax wedge, and explain why rebating every dollar does not eliminate the latter. [4]

Optional tool extension. Vary only $\tau \geq 0$ on a rational grid containing $0, 2$ and small positive taxes. Check both private budgets, government balance, both resource equations and the exact squared utility gain identity for the proposed Pareto improvement. Keep the rebate rule fixed and label $\tau = 0$ separately, where the strict gain becomes zero.

Problem 11.A7 [Advanced]. Observed production and the empirical content of general equilibrium.

Restored from the Western University Fall 2018 final examination. The two original tasks are retained. The supporting inequality indices, income accounting and fixed consumer identity are made explicit.

Suggested time: 60 minutes. Total: 60 points. You may use Afriat's theorem, but must explain how it is applied and how the market and ownership restrictions enter.

There are L goods, a finite set of market observations $t = 1, \ldots, T$, and the same finite set of consumers $i = 1, \ldots, I$ in each observation. Consumer i has one preference across all observations. Rationalization means that there exist continuous, concave and strictly increasing utilities $u_i : \mathbb{R}^L_+ \to \mathbb{R}$ for which the observed market conditions are competitive equilibria.

For every observation t, the data include positive prices $p^t \in \mathbb{R}^L_{++}$, positive individual incomes $b^t_i > 0$, aggregate endowment $\omega^t \in \mathbb{R}^L_+$, and the net production plan $y^t \in \mathbb{R}^L$ of the unique local firm. The common technology $Y \subseteq \mathbb{R}^L$ is known, nonempty, closed, convex and satisfies free disposal. Observed plans must belong to Y and maximize profit at their observed prices. Individual consumption bundles are unobserved. Individual nonnegative endowments summing to ω^t and nonnegative ownership shares summing to 1 may vary across observations; incomes include endowment value and the individual's share of firm profit. There are no unobserved outside transfers.

(a) Write a necessary and sufficient rationalization certificate using latent bundles x^t_i, utility levels U^t_i and positive multipliers λ^t_i. Include consumer budgets, all cross observation consumer inequalities, goods market clearing, firm profit maximization and aggregate income accounting. Prove sufficiency by constructing utilities from the certificate. Also show that the aggregate income identity allows a feasible allocation of individual endowments and ownership shares consistent with the stated incomes. Distinguish what is a finite latent consumer system from what must be checked against the known technology. [35]

(b) Now specialize to the same single consumer observed in every market and to the original free disposal technology $Y = \mathbb{R}^L_-$. The observed data are $(p^t, b^t, \omega^t)^T_{t=1}$. Prove that these data are rationalizable if and only if $b^t = p^t \cdot \omega^t$ in every observation and the price and bundle data $(p^t, \omega^t)^T_{t=1}$ satisfy GARP. State GARP precisely. Explain why the result would not be valid if each observation belonged to a different consumer with an unrelated preference. [25]

Part II: Formal and applied extensions

Problem 11.2 [Proof]. First Fundamental Welfare Theorem.

Consider a private-ownership economy with locally nonsatiated preferences and profit-maximizing firms.

(a) State the definition of Pareto efficiency used in the theorem.

(b) Prove that a competitive equilibrium allocation is Pareto efficient.

(c) Identify exactly where local nonsatiation, budget feasibility, profit maximization, and market clearing enter the proof.

Problem 11.3 [Proof]. Supporting a Pareto optimum.

Suppose consumption sets and production sets are closed and convex, preferences are continuous, convex, and locally nonsatiated, and a feasible allocation is Pareto efficient. Assume free disposal and that the aggregate net-resource-requirement set used below is closed, convex, and upward comprehensive. Assume the remaining separation hypotheses stated in the chapter.

(a) Formulate the aggregate set-separation argument and show how it supports the allocation as a price quasi-equilibrium with transfers.

(b) Explain why the initial endowment distribution generally must be changed by lump-sum transfers.

(c) State the positive-wealth or boundary condition needed to upgrade the quasi-equilibrium to a price equilibrium with transfers, and distinguish that object from a competitive equilibrium for the original ownership economy.

A consumer-owned firm

Chapter use case. Seed: Problems 11.1, 11.4. Fixed: all unlisted primitives and assumptions. Scaled: the production exponent from one-half to $\beta \in (0, 1)$. Tool: Codex + Python. Verification: Recover the square-root economy; verify planner and decentralized allocations coincide; check second-order conditions, profit finiteness, both market-clearing equations, and limiting behavior in β.

Problem 11.4 [Applied]. One consumer and one producer.

There are two goods. Good 1 can be consumed or used as input z to produce good 2 according to $y_2 = \sqrt{z}$. The consumer owns the firm and has an endowment $\bar{\omega}_1 > 0$ of good 1 and no good 2. Utility is $\log x_1 + \log x_2$.

(a) Solve the planner's problem.

(b) Find supporting relative prices from marginal conditions.

(c) Verify that the consumer and firm choices clear both markets.

11.2 Existence, uniqueness and the core

Part I: By hand

Problem 12.1 [Core]. Differential identities for excess demand.

Let aggregate excess demand $z : \mathbb{R}^L_{++} \to \mathbb{R}^L$ be continuously differentiable, homogeneous of degree zero, and satisfy Walras' law.

(a) Differentiate homogeneity to show $D_p z(p)p = 0$.

(b) Differentiate Walras' law to show $[D_p z(p)]^\top p = -z(p)$.

(c) Specialize the second identity to an equilibrium and interpret the left null vector.

Advanced exam practice

The suggested time applies to the hand questions, not the optional tool extension. Start with a diagram or solution plan, state the constraints, check all regimes and explain global optimality and economic accounting. Attempt the hand questions before using tools. These problems are additional preparation, not extra weekly requirements unless assigned in class. Before an optional extension, state Seed, Fixed, Scaled and Risk and enlarge one feature at a time.

Problem 12.A1 [Advanced]. One allocation and a whole interval of equilibrium prices.

Suggested time: 35 minutes for the hand part. Total: 50 points. Work by hand.

Two consumers have $u^A(x) = 2x_1 + x_2$ and $u^B(x) = x_1 + 2x_2$ on $\mathbb{R}^2_+$. Endowments are $\omega^A = (4, 0)$ and $\omega^B = (0, 4)$. Normalize prices to $(q, 1)$ with $q > 0$.

(a) Derive both full demand correspondences, including the prices at which a consumer is indifferent across a budget line. Find every normalized equilibrium price and every equilibrium allocation. [14]

(b) Give a KKT or direct budget certificate supporting the equilibrium allocation at every price you found. Why would imposing equality of the two marginal rates of substitution incorrectly reject the allocation? [10]

(c) Draw the entire Pareto set and determine the strong core under the book's strong blocking convention. Explain whether the original endowment permits any mutually acceptable strict gain from trade. [12]

(d) At an equilibrium price strictly inside the equilibrium price interval, calculate the derivative of normalized aggregate excess demand. Is this equilibrium regular? Distinguish uniqueness of an allocation, local uniqueness of normalized prices and the single valuedness of individual demand. [8]

(e) Does this example contradict the chapter's strict gross substitutes uniqueness proposition? Identify the missing strictness rather than treating linear utility as evidence of irrational behavior. [6]

Optional tool extension. Vary only $q > 0$ on a rational grid including $1/2, 2$, points between them and points just outside them. Implement full demand correspondences at the two equality prices. Verify the clearing selection and corner entry inequalities exactly, and check the reduced derivative inside the interval. Explain why sampling many clearing prices does not by itself prove the full interval claim.

Part II: Formal and applied extensions

Problem 12.2 [Proof]. Regularity and local uniqueness.

Normalize prices by $p_L = 1$ and let $\widehat{z}$ be the first $L - 1$ excess-demand equations.

(a) Define a regular equilibrium.

(b) Use the inverse function theorem to prove that a regular equilibrium is locally isolated.

(c) Suppose every normalized equilibrium is regular. Explain why compactness and boundary conditions then imply that there are only finitely many equilibria.

Problem 12.3 [Proof]. Gross substitutes and uniqueness.

Assume aggregate excess demand satisfies strict gross substitutes on the relevant price domain: increasing one price strictly raises excess demand for every other good whenever the comparison is nondegenerate. Suppose two strictly positive normalized equilibrium price vectors exist.

(a) Compare the largest relative price change across goods.

(b) Use gross substitutes and Walras' law to derive a contradiction unless the two price vectors are proportional.

(c) Explain why weak gross substitutes alone need not imply uniqueness and where strictness (or an irreducibility substitute) enters the proof.

A coalition challenges a competitive allocation

Chapter use case. Seed: Problem 12.4. Fixed: all unlisted primitives and assumptions. Scaled: the blocking predicate from all-strict improvement to weak improvement for all members and strict improvement for one witness. Tool: Codex + Lean. Verification: Compile the Lean strong-coalition price-support declaration and match each hypothesis to the economic proof. Lean verifies the finite price-summing contradiction, not the derivation of price support from Walrasian optimization or the replica-economy convergence claim. This case closes the price-support and core thread; Problems 12.2 and 12.3 are separate regularity and uniqueness extensions.

Problem 12.4 [Core]. Competitive allocations and the strong core.

Use the strong-blocking convention in a finite pure-exchange economy with locally nonsatiated preferences. A nonempty coalition S strongly blocks x when it can choose $(y_i)_{i\in S}$ such that $\sum_{i\in S} y_i = \sum_{i\in S} \omega_i$, $y_i \succeq_i x_i$ for every $i \in S$, and $y_j \succ_j x_j$ for at least one $j \in S$. Define the associated strong core.

(a) Prove that every Walrasian allocation belongs to the strong core.

(b) Give an economic reason why the converse can fail with finitely many consumers.

(c) State the content, but not a proof, of the core-convergence theorem for replica economies.

11.3 Empirical restrictions and equilibrium identification

Part I: By hand

Problem 13.1 [Core]. Equilibrium Afriat feasibility.

At each observation t, prices $p^t \gg 0$, individual incomes $(w_i^t)_i$, and aggregate endowment ω^t are observed, while individual allocations are latent.

(a) Introduce latent consumptions x_i^t and write individual budget equalities $p^t \cdot x_i^t = w_i^t$ and aggregate market clearing $\sum_i x_i^t = \omega^t$.

(b) Write Afriat inequalities for every consumer across observations.

(c) Explain why joint feasibility is a finite semialgebraic revealed-equilibrium test, why the latent multipliers and bundles make the full system bilinear, and how quantifier elimination projects the restrictions onto the observables.

Part II: Formal and applied extensions

Problem 13.2 [Proof]. What equilibrium data identify.

Suppose only equilibrium prices and aggregate endowments are observed.

(a) List the preference, allocation, and ownership objects that remain unobserved.

(b) Explain why market clearing alone cannot identify individual demand.

(c) Describe how additional observations, known individual endowments, or parametric restrictions can turn an underidentified model into a testable one.

A counterfactual equilibrium identified set

Chapter use case. Seed: Problems 13.1, 13.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of equilibrium observations from two to T, with the maintained preference class and counterfactual fixed. Tool: Codex + Python. Verification: Verify feasibility and market clearing for every retained candidate, show how each added observation changes the bounds, and never replace set identification by a point estimate.

Problem 13.3 [Applied]. Counterfactual discipline.

A dataset passes a revealed-equilibrium feasibility test under a maintained class of preferences.

(a) Explain why passing the test does not point-identify a counterfactual equilibrium.

(b) Define an identified set for a counterfactual price or allocation.

(c) State two economically meaningful restrictions that could tighten this set, and explain the cost of each restriction.

Part V

Advanced Equilibrium Applications

Chapter 12

Niche Consumption and Product Space

12.1 Reading and existing practice

Read Chapter 12 of the notes for the evidence on niche consumption, the geometry of differentiated products, neighborhood substitution and bundles represented by measures. This is a research synthesis with no additional weekly assignment. The references below revisit existing problems; they do not add questions or change the weekly plan.

Problems 3.1 and 3.3 review finite revealed preference comparisons and Afriat certificates. They provide the finite benchmark for the chapter's continuum argument, not a proof that every finite result extends to an arbitrary product space. Problem 6.1 isolates the logit restriction on substitution through its odds ratios and IIA.

Problems 8.1 and 8.2 state sufficient conditions for exact aggregation. Use those conditions to distinguish an individual household's basket from aggregate demand. Problem 11.A4 separates redistribution from a change in aggregate resources; Problem 13.1 combines individual optimality across observations with market clearing.

Problem 15.1 introduces bilateral stability through deferred acceptance. Compare that allocation problem with choosing a nearby product, but do not identify geometric proximity with stability when both sides have preferences and partners have limited capacity.

Chapter 13

Sequential Trade

Part I: By hand

Problem 14.1 [Core]. Spanning contingent consumption.

There are S states tomorrow and K traded assets with payoff matrix $R \in \mathbb{R}^{S \times K}$ and date-zero asset-price vector $q \in \mathbb{R}^K$. Portfolios are unrestricted, $\theta \in \mathbb{R}^K$, so short sales are allowed. Assume no arbitrage.

(a) State the rank condition under which every contingent net trade can be implemented by an asset portfolio.

(b) If $S = 2$ and the assets have payoff vectors $(1, 1)$ and $(1, 0)$, construct the portfolio that delivers any target payoff (c_1, c_2).

(c) Let $\psi \in \mathbb{R}^S_{++}$ be state prices satisfying $q = R^\top \psi$. Explain how the date-zero portfolio budget maps into an Arrow-Debreu present-value budget.

Problem 14.2 [Core]. Incomplete markets.

Let $S = 3$ and suppose the only assets are a bond with payoff $(1, 1, 1)$ and an asset with payoff $(1, 0, 0)$.

(a) Characterize the attainable payoff subspace.

(b) Determine whether the claim $(0, 1, -1)$ is attainable.

(c) Explain why competitive equilibrium with these assets need not be Pareto efficient relative to the full contingent-commodity economy.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

Complete versus incomplete sequential markets

Chapter use case. Seed: Problems 14.1, 14.2. Fixed: all unlisted primitives and assumptions. Scaled: the asset span from rank $S - 1$ to rank S at a fixed number of states. Tool: Codex + Python; Lean target. Verification: Check portfolio replication, the state-price units map, and

present-value budgets; include a rank-deficient claim that cannot be implemented. Arrow-Radner equivalence remains a future Lean target.

Problem 14.3 [Proof]. Arrow-Debreu and sequential equivalence.

There are S states, K assets with payoff matrix $R \in \mathbb{R}^{S\times K}$ of rank S, date-zero asset prices q, and state-s spot-price vectors p_s. Asset payoffs are denominated in the state numeraire. Consumer i chooses a portfolio z_i subject to $q\cdot z_i \leq 0$, then chooses x_{is} subject to $p_s\cdot x_{is} \leq p_s\cdot\omega_{is}+p_{s1}(Rz_i)_s$. Assume no arbitrage and strictly monotone continuous preferences.

(a) Map an Arrow-Debreu budget into date-zero asset purchases and state-by-state spot trades.

(b) Prove the reverse mapping from a sequential budget to a present-value budget.

(c) Identify which step fails under incomplete markets or binding portfolio constraints.

Chapter 14

Matching

Part I: By hand

Problem 15.1 [Core]. Deferred acceptance by hand.

Three men have rankings $m_1 : w_1 \succ w_2 \succ w_3$, $m_2 : w_1 \succ w_2 \succ w_3$, and $m_3 : w_2 \succ w_3 \succ w_1$. The women rank $w_1 : m_2 \succ m_1 \succ m_3$, $w_2 : m_1 \succ m_3 \succ m_2$, and $w_3 : m_3 \succ m_2 \succ m_1$.

(a) Run men-proposing deferred acceptance round by round.

(b) Verify stability of the resulting matching by checking potential blocking pairs.

(c) Explain why termination is guaranteed in every finite market.

Part II: Formal and applied extensions

Problem 15.2 [Proof]. Proposer optimality.

Consider a finite balanced one-to-one matching market in which every participant has a complete strict ranking of the other side and all partners are acceptable.

(a) Prove that if a receiver rejects a proposer during deferred acceptance, that rejected proposer-receiver pair cannot be matched to each other in any stable matching.

(b) Use this rejection lemma to prove proposer optimality among stable matchings.

(c) Explain why the receiver side obtains its least-preferred stable matching.

Housing allocation and coalitional stability

Chapter use case. Seed: Problem 15.3. Fixed: all unlisted primitives and assumptions. Scaled: the number of agents and houses from three to n. Tool: Codex + Python; Lean target. Verification: Recover the three-agent top-trading-cycles allocation; check permutation, individual rationality, Pareto comparisons, and coalition blocking by enumeration. Top trading cycles and housing-core claims remain future Lean targets.

Problem 15.3 [Core]. Top trading cycles.

Agents $1, 2, 3$ initially own houses h_1, h_2, h_3. Their rankings are $1 : h_2 \succ h_1 \succ h_3$, $2 : h_1 \succ h_2 \succ h_3$, and $3 : h_1 \succ h_3 \succ h_2$.

(a) Run top trading cycles.

(b) Show that the outcome is individually rational and Pareto efficient.

(c) Verify directly that no coalition blocks the outcome.

Chapter 15

Computable General Equilibrium

Part I: By hand

Problem 16.1 [Core]. Calibrating a CES technology.

A constant-returns technology is $Q = A[\alpha K^{-\rho} + (1-\alpha)n^{-\rho}]^{-1/\rho}$, with $\rho > -1$ and $\rho \neq 0$. A strictly positive interior cost-minimizing benchmark observation gives (Q, K, n, r, W) and an elasticity of substitution $\sigma > 0$, $\sigma \neq 1$. (The Cobb-Douglas limit $\sigma = 1$ is outside the displayed parameterization.)

(a) Recover ρ from σ.

(b) Use the cost-minimizing first-order condition to recover α.

(c) Recover A and verify that the benchmark is exactly replicated.

Problem 16.2 [Core]. Social-accounting consistency.

A two-sector social accounting matrix records sectoral sales, intermediate purchases, labor and capital payments, household income, taxes, savings, and imports.

(a) State the row-balance and column-balance conditions.

(b) Explain why every market-clearing equation has a corresponding accounting identity but not every accounting identity is a behavioral equation.

(c) Design a numerical residual check that would detect an incorrectly calibrated benchmark.

Advanced exam practice

The suggested time applies to the hand questions, not the optional tool extension. Start with a diagram or solution plan, state the constraints, check all regimes and explain global optimality and economic accounting. Attempt the hand questions before using tools. These problems are additional preparation, not extra weekly requirements unless assigned in class. Before an optional extension, state Seed, Fixed, Scaled and Risk and enlarge one feature at a time.

Advanced case guide. Problems 16.A1 and 16.A2 restore both the theory and Arabasta and Baltigo modeling parts of the SFU 2024 final examination. Problem 16.A3 is a new general equilibrium and dynamic extension of the Galor household problem 1.6. Each separates hand benchmarks from computational verification and states its equilibrium closure explicitly. Separate student packets contain these same questions.

Problem 16.A1 [Advanced]. CES trade, market clearing and the limits of a fixed point argument.

Restored from the theory problem in the SFU Fall 2024 final examination. The original four stage argument and CES weighting are retained. The symmetry and fixed point claims are corrected. The following timing and points are for practice, not the original exam allocation.

Suggested time: 70 minutes. Total: 60 points for Part I.

Part I. By hand

There are $N \geq 2$ countries, indexed by i for origins and j for destinations. Country i owns a fixed quantity $Q_i > 0$ of its unique variety, has no other goods endowment, and has factory price $p_i > 0$. Its income is $y_i = p_i Q_i$. All varieties are traded competitively. Delivering one unit from i to j uses $t_{ij} \geq 1$ origin units, with $t_{ii} = 1$. Delivered prices are $p_{ij} = p_i t_{ij}$. There are no outside transfers or trade deficits. Destination j chooses nonnegative delivered quantities c_{ij} with budget $\sum_i p_i t_{ij} c_{ij} \leq y_j$ and utility $U_j(c) = [\sum_i \beta_i^{(1-\sigma)/\sigma} c_{ij}^{(\sigma-1)/\sigma}]^{\sigma/(\sigma-1)}$, where $\beta_i > 0$ and $\sigma > 0$, $\sigma \neq 1$. For $0 < \sigma < 1$, define utility to be zero when any component is zero, its continuous boundary value. You may use the standard concavity of this CES aggregator. Bilateral expenditure is $x_{ij} = p_i t_{ij} c_{ij}$, including domestic expenditure when $i = j$. Let $Y = \sum_i y_i$ and $\theta_i = y_i / Y$.

(a) Derive the optimal bilateral quantities, bilateral expenditures and destination unit expenditure index P_j, step by step. Explain why the optimum exhausts the budget, is interior and is globally optimal. Your argument must preserve the displayed β_i weighting. [16]

(b) State physical market clearing, including iceberg use, and its nominal form. Use your expenditure formula to isolate $\beta_i p_i$ as a function of y_i, the trade costs and the destination indices. Explain why the income identity $y_i = p_i Q_i$ must still hold when interpreting the result as the original fixed supply economy. [10]

(c) Define outward multilateral resistance by $\Pi_i^{1-\sigma} = \sum_j \theta_j (t_{ij}/P_j)^{1-\sigma}$. Eliminate $\beta_i p_i$ to derive the structural gravity equation and the inward resistance system. Show the reciprocal scale freedom of the two resistance systems and state a normalization. Take logs to identify the exporter effect, importer effect and coefficient on $\log t_{ij}$. Distinguish the conditional index normalization from changing nominal currency units in the underlying economy. [16]

(d) Suppose $t_{ij} = t_{ji}$. You may use the following result without proof: for a strictly positive finite trade cost matrix and positive shares summing to one, the joint inward and outward resistance system has a positive index pair, and any two such pairs differ only by reciprocal scalar rescaling. Use this result to explain when a normalization with $\Pi_i = P_i$ is available, and why symmetry alone does not make every unnormalized pair equal.

For the rest of this part use exactly the original diagnostic values $N = 2$, $\sigma = 1/2$, $t_{ij} = 1$ and $\theta_i = 1/2$. After choosing the symmetric normalization, write the inward equations as $P = g(P)$. Determine whether g maps $\{P : P_i \geq 1\}$ into itself. Then use $K = [1/2, 2]^2$: prove that g maps K into itself and apply Brouwer to establish a fixed point. Determine all positive fixed points directly. Compute Dg at $(1, 1)$ and decide whether an induced matrix norm can be strictly below one there. Finally calculate two iterations from $P^{(0)} = (2, 2)$ and explain what they show about repeated substitution, existence and uniqueness. [18]

Part II. Codex and verification

Construct a synthetic economy with at least ten countries, positive β_i, Q_i, finite iceberg costs and $\sigma > 1$. Keep fixed physical supplies rather than silently treating both factory prices and nominal incomes as independent data. Solve for positive factory prices in log variables with one numeraire normalization. Compute incomes, delivered consumption, expenditure, both resistance systems and the physical shipment totals. Check every consumer budget, both row and column expenditure sums, all physical goods markets, positivity and the omitted redundant equation. Repeat under a change of currency units and verify how prices, incomes, nominal trade and physical quantities transform. Compare a normalized resistance iteration with a root solver conditional on the resulting income shares, align their index gauges before comparing them, and report full residuals as well as stopping criteria. For the synthetic fixed effect regression, require nonadditive bilateral variation in $\log t_{ij}$ and report the design rank after choosing a dummy normalization. Recover the theoretical cost coefficient. Reproduce the two country diagnostic separately and do not infer convergence from an equation having a unique solution.

Problem 16.A2 [Advanced]. Arabasta and Baltigo: building the original gravity model of trade.

Restored from Problem 2 of the SFU December 2024 final examination. The original six modeling tasks, Arabasta and Baltigo, unequal country incomes, transport technology and distance comparisons are retained. Part II is a new computational extension.

Part I. By hand

Suggested time: 75 minutes. Total: 60 points.

There are two countries, Arabasta a and Baltigo b, and one good from each origin. Quantity c_{ij} is consumption in destination j of origin i's good. Nominal income and expenditure are $y_a, y_b > 0$. Origin prices are $p_a, p_b > 0$. Domestic delivery has no resource loss, while bilateral trade has the symmetric iceberg factor $\tau \geq 1$: $p_{aa} = p_a$, $p_{bb} = p_b$, $p_{ab} = p_a\tau$ and $p_{ba} = p_b\tau$. Delivering one unit abroad requires τ units at origin. Bilateral expenditure is $x_{ij} = p_{ij}c_{ij}$ and origin revenue clearing requires $y_i = x_{ia} + x_{ib}$.

Use equal origin weights and common CES preferences. Write the substitution elasticity as $\sigma > 1$ and use $\sigma = 2$ for the closed form equilibrium calculations. The original key's power utility with exponent $1/2$ is ordinally equivalent to this case; its exponent is not the substitution elasticity.

Normalize $p_b = 1$. In Part I, y_a and y_b are fixed nominal incomes measured in these units. Origin supply quantities are not independently fixed: after solving the revenue equations they are implied by $Q_i = y_i/p_i$. State this conditional nominal income closure when interpreting a counterfactual. It is not a comparison that holds both nominal incomes and physical endowments fixed.

(a) Original task 1.1. Specify a one parameter CES utility family and Arabasta's budget. Explain which goods, prices and income enter this problem. [10]

(b) Original task 1.2. Write the Lagrangian and derive both consumption quantities in Arabasta. Relabel carefully to obtain Baltigo's demands. Give a global optimality argument and state where your solution is interior. [10]

(c) Original task 2.1. Write both countries' revenue clearing conditions. Specialize to $\sigma = 2$, impose $p_b = 1$, and solve explicitly for the unique positive origin price p_a as a function of y_a, y_b, τ. Reject any algebraic root that is not an admissible price, and check the case $\tau = 1$ directly. [10]

(d) Original task 2.2. Let $d \geq 0$ be distance. Propose a one parameter engineering transport technology $d = f(k)$ with nonincreasing returns in ship capital services k. Normalize the rental cost of these services to one in the origin good resource units used to define iceberg shipping. Derive the per delivered unit resource requirement $C(d)$ and set $\tau(d) = 1 + C(d)$. Verify $\tau(0) = 1$ and that larger distance increases the barrier. Explain why you must not add a second monetary transport bill to the country budget. The historical key's constant returns choice is admissible; a concave power technology is also admissible. [10]

(e) Original task 3.1. Eliminate the origin price from Arabasta's equilibrium import quantity c_{ba} and nominal imports x_{ba}. Derive the corresponding exports c_{ab}, x_{ab} and verify both budgets and both physical resource equations, including iceberg use. As a common numerical check, use $y_a = 5$, $y_b = 2$, $d = 1$ and the admissible technology $f(k) = \sqrt{k}$. This check is part of the existing task, not an additional scored question. [10]

(f) Original task 3.2. Prove that doubling both nominal incomes doubles bilateral nominal trade in this closure. Then prove that increasing distance strictly reduces bilateral nominal imports, holding those incomes fixed. Separate the response of import quantities from nominal import values, and explain why this is not yet a welfare theorem for a fixed physical endowment economy. [10]

Part II. Codex and verification

This is a new extension, separate from the restored final question. Begin only after completing the hand arguments.

1. Use the numerical seed from part (e), fix the engineering parameter, and vary only distance over $d \geq 0$. Ask Codex to implement your analytical price, expenditure and quantity formulas, and independently solve the two revenue equations. Compare both methods. Test positivity, budget exhaustion, balanced bilateral expenditure, the domestic expenditure residuals, and every origin resource equation. Include zero distance without evaluating a formula that divides by zero. Plot bilateral expenditure and both implied origin supply quantities. Explain explicitly which physical quantity changes when nominal GDP is held fixed.

2. Optional different closure. Freeze physical endowments at the values you derived from the seed, and retain $p_b = 1$. Now income must be endogenous, $y_a = p_a Q_a$ and $y_b = Q_b$. Derive the positive price equation under this closure, establish uniqueness of its positive root, and solve it as distance varies. Verify the original consumer problems, physical market clearing and income accounting independently. Compare the two closures at the seed and away from it. Any utility comparison must use the same CES preference and the full fixed endowment closure, not interpret the fixed nominal GDP family as the same physical economy.

Problem 16.A3 [Advanced]. From household KKT to a land constrained growth economy.

Suggested time: 90 minutes for the hand benchmarks. Total: 80 points. The optional program is not included in this time. This teaching case extends Problem 1.6 to competitive factor prices and a sequence of generations. It is inspired by Galor's quantity versus quality mechanism, not a

reproduction of his full Unified Growth Theory.

There is one consumption good, whose price is one, fixed land $X > 0$, and a mass $N > 0$ of identical current adults with skill $h > 0$. A competitive constant returns firm rents land $Z \geq 0$ and raw adult labor hours $L \geq 0$ to produce $Y = AZ^{1-\alpha}(hL)^{\alpha}$, where $A > 0$ and $0 < \alpha < 1$. The land market must clear at $Z = X$. The wage $w > 0$ is paid per raw hour, and R is rent per unit of land. Every adult owns an equal land share and receives $\rho = RX/N$. Firm profit is distributed but is zero at a competitive supporting price.

Each adult has one unit of time. It chooses consumption $c > 0$, continuous fertility $n > 0$ and education time per child $e \geq 0$ to maximize $(1-\gamma)\log c + \gamma \log[nh_a(e)]$, where $0 < \gamma < 1$, $\tau > 0$, $a > 0$, $\overline{c} > 0$ and $h_a(e) = 1 + ae/(\tau + e)$. Total child time is $t = n(\tau + e)$. Constraints are $c + wt \leq w + \rho$, $t \leq 1$ and $c \geq \overline{c}$. Land rent is nonlabor income: child time sacrifices wages, not rent. Each household takes w and ρ as given. The logarithm excludes zero fertility in this model.

Part I. By hand

(a) Set $E = ne$ and $Q(n, E) = n + anE/(\tau n + E)$. State the transformed feasible set and write the complete KKT system, including the time bound, subsistence and the zero education entry inequality. Establish concavity and explain why a feasible KKT certificate gives a global optimum. [12]

(b) At fixed prices, write $I = w + \rho > \overline{c}$. Solve globally for education, child time and consumption, including all three possible bounds on child time and the equality case $a = 1$. Explain why replacing w by I in the cost of child time would be an economic error. [12]

(c) Define $B = Ah^{\alpha}(X/N)^{1-\alpha}$. A symmetric general equilibrium has $L = N(1-t)$ and $Nc = Y$. Derive factor prices, zero profit and the resource identity. Find all positive equilibria for every $B > \overline{c}$, identify the subsistence threshold and verify the household KKT certificate at the equilibrium. Explain what happens to feasibility when $B \leq \overline{c}$. [16]

Dynamic benchmarks. Continue by hand. The optional program at the end checks these calculations; it does not replace the arguments. At the end of a generation all children become next generation adults, so $N' = nN$. Skill is inherited and augmented by education: $h' = hh_a(e)$. Land and A stay fixed, and land ownership is shared equally among the next cohort. Parents value child quantity and quality through the displayed utility but do not solve a forward looking dynamic program. Thus $n = 1$ means replacement of the current adult mass, not a literal demographic birth rate.

(d) Derive the one dimensional law for B'. Set $\alpha = 3/4$, $\gamma = 1/2$, $\tau = 1/4$ and $\overline{c} = 4$. With $a = 1/2$, find a subsistence steady state with replacement fertility and constant skill. Calculate the derivative of the local B map there and explain the stability conclusion, including its scope. [12]

(e) Start at that steady state and permanently raise a to 3. Calculate the impact education, child time, fertility and skill multiplier. Does the next generation cross the threshold into slack subsistence? Prove your comparison exactly. Compute the subsequent fertility and asymptotic growth factors of skill, adult mass, B and consumption. [18]

(f) Explain which feedbacks close the economy, why rents alter fertility incentives without being sacrificed per hour, and why the model does not deliver exact zero fertility or a full demographic transition. Distinguish an exogenous education return shock from endogenous technological progress, and distinguish the impact fertility response from its later level. [10]

Part II. Codex and verification

Optional tool extension. Keep $A = 8$, $X = h_0 = 1$ and all calibrated parameters fixed, with the permanent $a = 1/2$ to $a = 3$ change at date zero. Vary only initial adult mass $N_0 \in (0, 16)$, including the steady state and regime boundary populations obtained by hand and points immediately around them. State Seed, Fixed, Scaled and Risk before coding. Simulate twenty generations using independently implemented household and equilibrium routines. Check all budgets, time and subsistence bounds, raw labor and land clearing, zero profit, aggregate goods clearing, KKT signs and complementary products at every date. Compare the household formula with an independent bounded optimization at selected dates and state tolerances. Plot the paths and mark the subsistence switch. Report what is proved analytically, what is checked on this finite grid and what is only a maintained demographic assumption.

Part II: Formal and applied extensions

This chapter moves directly from the hand benchmark to its chapter use case.

A tariff counterfactual from partial to general equilibrium

Chapter use case. Seed: Problems 16.1, 16.2, 16.3. Fixed: all unlisted primitives and assumptions. Scaled: model closure from fixed factor prices to one endogenous shared labor market linking import-competing and export sectors. Tool: Codex + Python. Verification: Replicate the partial-equilibrium benchmark, then report factor-market, goods-market, trade-balance, government-budget, and welfare residuals. Identify which change comes from the single added general-equilibrium channel.

Problem 16.3 [Applied]. A tariff counterfactual.

A small open economy takes the world price P^W and exchange rate ER as given. A tariff τ implies domestic price $P = \mathrm{ER}\, P^W(1+\tau)$. Let differentiable domestic demand and supply be $D(P)$ and $S(P)$, imports be $M(P) = D(P) - S(P) > 0$, and tariff revenue be returned lump-sum, $\mathrm{TR}(\tau) = \tau\, \mathrm{ER}\, P^W M(P(\tau))$. Interpret D as quasi-linear (or otherwise free of income effects) and S as competitive supply generated by marginal cost, so consumer and producer surplus are valid partial-equilibrium welfare measures.

(a) Derive the first-order change in the domestic price from a small tariff change.

(b) Derive the symbolic first-order decomposition $d\mathcal{W} = d\,\mathrm{CS} + d\,\mathrm{PS} + d\,\mathrm{TR}$ and simplify it at free trade. Do not assume numerical functional forms.

(c) Explain which general-equilibrium channels are omitted by that approximation and must be restored in a CGE counterfactual.

Appendix A

Order Theory

Part I: By hand

Problem A.1 [Core]. Classifying relations.

Assume $|X| \geq 2$. For each relation below, determine whether it is reflexive, irreflexive, symmetric, asymmetric, antisymmetric, transitive, and complete: equality on X; strict set inclusion on 2^X; weak set inclusion on 2^X; and the componentwise order on $\mathbb{R}^2$.

Give a counterexample for every failed property.

Part II: Formal and applied extensions

Problem A.2 [Proof]. Quotienting a preorder.

Let R be a preorder on X and define xI_Ry if and only if xRy and yRx.

(a) Prove that I_R is an equivalence relation.

(b) Define $[x]\,\bar{R}\,[y]$ if and only if xRy. Prove that this definition is independent of representatives.

(c) Prove that $\bar{R}$ is a partial order on X/I_R.

Problem A.3 [Proof]. Maximal, greatest, and finite existence.

Let R be a preorder on a nonempty finite set S.

(a) Prove that S has an R-maximal element.

(b) Give an example with several maximal elements and no greatest element.

(c) Prove that if R is complete, maximal and greatest elements coincide.

Finite preference completion and multiple utilities

Chapter use case. Seed: Problems A.1, A.2, A.3. Fixed: all unlisted primitives and assumptions. Scaled: the number $|X|$ of alternatives in a finite preorder, beginning with the hand example. Tool: Codex + Lean. Verification: Verify the preorder, quotient construction, and each proposed linear extension. Szpilrajn extension is currently verified; quotient pullback, intersection of all extensions, and the finite multiple-utility corollary remain explicit formal targets.

Problem A.4 [Proof]. Order extensions and multiple representation.

Let R be a preorder on a finite set X.

(a) Pass to the quotient partial order on X/I_R and extend it to a linear order.

(b) Pull the extension back to a complete preorder on X.

(c) Show that R is the intersection of its complete preorder extensions and deduce a finite multiple-utility representation.

Additional Practice Reserve

These ten question-only problems are corrected and independently checked descendants of earlier course assessments. Each combines a hand-solvable benchmark with a controlled computational or formal scale-up.

A.1 Revealed Preference and Recoverability

Problem R001 [Proof]. Quasilinear Afriat inequalities and welfare bounds.

Suggested weekly placement: PS03.

Let $\mathcal{O} = \{(p^t, x^t)\}_{t=1}^{T}$, where $p^t \gg 0$ and $x^t \in \mathbb{R}^L_+$.

Part I -- by hand. Prove that the observations can satisfy $x^t \in \arg\max_{x \in \mathbb{R}^L_+}\{u(x) - p^t \cdot x\}$ only if there are numbers $v^1, \ldots, v^T$ such that $v^t - v^s \geq p^t \cdot (x^t - x^s)$ for every s, t. Prove the converse by studying $u(x) = \min_t\{v^t + p^t \cdot (x - x^t)\}$. Establish continuity, concavity, strict monotonicity, and rationalization rather than merely citing Afriat's theorem.

For $V(p) = \max_{x \in \mathbb{R}^L_+}\{u(x) - p \cdot x\}$, suppose optimizers $x(p)$ and $x(q)$ exist. Derive upper and lower bounds on $V(p) - V(q)$ using the two observed demands. Extend the argument to a finite path of price vectors and state precisely when a componentwise price increase gives a strict welfare loss.

Part II -- verified scale-up. Express the inequalities as a difference-constraints graph. With Codex and Python, implement a feasibility test for arbitrary T and L, return one normalized vector v, and report a violating positive cycle when the system is infeasible. Unit-test the code on a small rationalizable data set and on a perturbed data set. Verify every returned inequality directly. As an optional Lean task, formalize the telescoping path bound conditional on the finite system of inequalities.

A.2 Abstract Choice and Order Foundations

Problem R002 [Proof]. Sequential satisficing with a default rule.

Suggested weekly placement: PS04.

Let X be finite, let $\triangleright$ be a strict linear (total) search order, and let $S = \{x : u(x) \geq \bar{u}\}$. From a nonempty menu A, inspect alternatives in the order $\triangleright$ and choose the first member of $A \cap S$. If $A \cap S$ is empty, choose the $\triangleright$-first member of A.

Part I -- by hand. Construct a single strict priority order $\succ^*$ whose maximal element agrees with this procedure on every nonempty menu. Use that representation to prove WARP

and SARP. Explain exactly why the fallback clause is necessary for a full-domain choice function. Give a three-alternative example and list the choice from every nonempty menu.

Part II -- verified scale-up. Encode finite menus, the search order, the threshold set, and the fallback choice. Ask Codex to generate either a Lean proof or an exhaustive checker establishing equality between sequential choice and $\succ^*$-maximization. Scale from three alternatives to all menus of a user-chosen finite set, and include a deliberately broken implementation without the fallback to exhibit its failure of full-domain nonemptiness.

Problem R003 [Applied]. Choice overload and attraction through menu-dependent attention.

Suggested weekly placement: PS06.

An outside option o is always considered. In a menu with n active products, each active product is independently considered with baseline probability $q_n = 1/n^2$; the consumer chooses the highest-u considered item, or o if none is considered. Utilities are strict. When a target b and its dominated decoy b^- are both offered, the consideration probability of b rises by $\delta = 1/3$, while all other baseline probabilities remain unchanged.

Part I -- by hand. Compare $A_2 = \{a, c, o\}$ with $A_3 = \{a, c, d, o\}$ and calculate the probability of choosing o. Next let $u(b) > u(c) > u(b^-) > u(o)$ and compare $A = \{b, c, o\}$ with $B = \{b, b^-, c, o\}$. Calculate the probability of choosing b in both menus. Prove that the resulting attraction inequality cannot be generated by any menu-independent random-utility model.

Part II -- verified scale-up. Build an exact enumerator over consideration sets and a Monte Carlo simulation. Confirm the hand calculations, then vary n, δ, and the ranking. Ask Codex to produce automated checks for probability adding-up, the parameter domain, regularity violations, and convergence of the simulation to the exact probabilities. Report which changes generate overload, attraction, both, or neither.

Problem R004 [Proof]. Maximal choice under incompleteness and a GARP failure.

Suggested weekly placement: PS04.

Let $X = \{a, b, c\}$. Define an incomplete preorder R by reflexivity together with bPa; alternative c is incomparable with both a and b. From each nonempty menu A, let $C(A)$ contain all R-maximal elements.

Part I -- by hand. Compute $C(A)$ for every nonempty menu. Define direct revealed preference by xR^Dy when some menu chooses x while making y feasible, and define xP^Dy when that menu rejects y. Show that the transitive closure $(R^D)^*$, together with P^D, violates the standard GARP test; identify an explicit witness. Explain why this does not contradict maximality under an incomplete preorder.

Part II -- verified scale-up. Represent finite preorders and maximal-choice correspondences in Lean or Python. Verify reflexivity and transitivity of the primitive relation, compute all maximal sets, and mechanically exhibit the revealed-preference path. Then enumerate all three-alternative preorders and classify which induced maximal-choice data satisfy GARP. Distinguish a failure of the data test from a failure of the primitive preorder axioms.

A.3 Production and Competitive Markets

Problem R005 [Proof]. Aggregating heterogeneous Cobb--Douglas firms.

Suggested weekly placement: PS08.

Firm j has technology $q_j = A_j \ell_j^{a_j} k_j^{b_j}$ with $A_j > 0$, $a_j, b_j > 0$, and $a_j + b_j < 1$. Its net-output set is $Y_j = \{(-\ell, -k, q) : q \leq A_j \ell^{a_j} k^{b_j},\ \ell, k \geq 0\}$. Output price is $p > 0$ and input prices are $w_\ell, w_k > 0$.

Part I -- by hand. Derive the unique profit-maximizing output, both input demands, and profit. State where decreasing returns are used. Prove that the aggregate technology is the Minkowski sum $Y = Y_1 + \cdots + Y_J$ and that its profit function is the sum of the firm profit functions. State the corresponding Hotelling result both at differentiable prices and in subdifferential form, using derivatives $(q_j, -\ell_j, -k_j)$ with respect to (p, w_ℓ, w_k).

Part II -- verified scale-up. Generate a heterogeneous population of firms and compute aggregate supply and factor demands. Verify numerically that direct maximization over firm-level allocations agrees with the sum of closed-form firm expressions. Use automatic or symbolic differentiation to check Hotelling's lemma, and test how the aggregate elasticity changes with the distribution of returns-to-scale parameters.

Problem R006 [Applied]. Selection versus capacity destruction in partial equilibrium.

Suggested weekly placement: PS09.

This is the distributional extension of problem 10.A5, the restored Western 2019 final examination question. Use its consumer preferences, unrestricted numeraire budget and producer cost. Here allow any consumer mass $\eta > 0$. Retain its distinction between actual removal and replacement of the producer population.

Part I: by hand. Complete problem 10.A5 first. Reproduce its uniform benchmark, both producer selection experiments, consumption variance and fixed initial wealth comparison as verification targets for the extension below. This reference replaces the previously duplicated uniform prompt; the complete five part question is now in the main chapter bank.

Part II: verified extension. Replace the uniform type distributions by beta distributions and allow an arbitrary survival cutoff. Define each probability law, the actual population measures and their masses before integrating. Derive the market clearing equation, solve it numerically over a parameter grid, and separate the effect of changing the conditional productivity distribution from changing producer mass. Include monotonicity checks and compare numerical integration with the uniform closed form. Hold initial wealth fixed in welfare comparisons and distinguish per consumer moments from population totals.

A.4 Equilibrium, Trade, and Computation

Problem R007 [Applied]. CES structural gravity and multilateral resistance.

Suggested weekly placement: PS11.

This is a companion to the full restored gravity exercise 16.A1, not a second copy of its derivation. For comparison with the earlier reserve notation, set $a_i = \beta_i^{1-\sigma}$ and $\tau_{ij} = t_{ij}$, with $\sigma > 1$. In this conditional resistance exercise, nominal incomes and shares are given; it does not independently solve the fixed physical supply economy.

Part I: by hand. Complete parts (a) to (c) of 16.A1. Translate the resulting equations into the a_i notation and use them as verification targets. Keep the distinction between origin revenue, destination expenditure and iceberg shipments.

Part II: verified extension. Begin with the same country set and income shares as your 16.A1 calculation. Vary only asymmetry in the finite positive trade cost matrix, retaining domestic cost one and all trade factors at least one. Solve both normalized resistance systems. Verify every row and column expenditure sum, positivity and reciprocal index scaling. Compare a normalized iteration with a root solver and report residuals, not only step changes. For synthetic fixed effect recovery, check design rank and include bilateral cost variation that is not absorbed by origin and destination effects.

Problem R008 [Proof]. Why a Banach argument fails even when existence and uniqueness hold.

Suggested weekly placement: PS10.

Use the map g and rectangle $K = [1/2, 2]^2$ from 16.A1(d). The full corrected hand diagnostic is now in the main chapter bank; this reserve entry supplies its numerical stability extension.

Part I: by hand. Complete the fixed point diagnostic in 16.A1(d). Retain its exact fixed point and cycle as verification targets. For the damped rule below, derive the local eigenvalues at the fixed point and distinguish local stability from a contraction bound over the whole rectangle.

Part II -- verified scale-up. Use interval arithmetic or Lean-supported inequalities to certify invariance of K and the derivative calculation. Compare plain iteration, the damped rule $P^{n+1} = (1 - \lambda)P^n + \lambda g(P^n)$ for $\lambda = 1/2$, and a root solver from multiple starting values. Ask Codex to create tests that distinguish existence, uniqueness, local stability, and global contraction; the report must not infer one property from another.

Problem R009 [Proof]. A regular two-good economy with three equilibria.

Suggested weekly placement: PS10.

Normalize prices to $(q, 1)$. Consumer 1 chooses $a \geq 0, b > 0$, has $u_1(a, b) = a - \frac{1}{8}b^{-8}$, and owns $(2, r)$. Consumer 2 chooses $c > 0, d \geq 0$, has $u_2(c, d) = -\frac{1}{8}c^{-8} + d$, and owns $(r, 2)$, where $r = 2^{8/9} - 2^{1/9}$.

Part I -- by hand. Derive both consumers' interior demands and prove that $a(q), d(q) > 0$ for every $q > 0$, so boundary demand cannot create an omitted equilibrium. Derive the good-2 excess-demand function. Show that $q \in \{1/2, 1, 2\}$ are equilibria and that there are no other positive normalized equilibria. Verify that each equilibrium is regular and locally isolated. Compute $z_2'(1)$ and compare it with the gross-substitutes requirement $z_2'(1) \geq 0$. Explain why regularity does not imply uniqueness.

Part II -- verified scale-up. Put $t = q^{1/9}$ and use a computer algebra system to certify the positive-root count without relying only on a plot. Check all equilibrium residuals and derivatives at high precision. Choose one scalar endowment perturbation, initialize continuation at each baseline equilibrium, and trace each local branch only until regularity fails or the declared small parameter interval ends. Connect the local numerical result to the implicit-function theorem. Optionally formalize the exact three roots and their nonzero derivatives in Lean.

Problem R010 [Applied]. A complete-markets Radner equilibrium by hand.

Suggested weekly placement: PS11.

There are two consumers, two states A, B, and two goods in each state. Both consumers have equal state probabilities and state utility $\log c_1 + \log c_2$. Each owns $\omega > 0$ units of good 2 in each state. Consumer 1's good-1 endowment is $(\omega/2, 3\omega/2)$ across states, and consumer 2's is reversed. Good 1 is the state numeraire and r_s is the state-s price of good 2. Arrow security s pays one unit of good 1 in state s. Consumer i holds θ_{is}, securities are in zero net supply, and the date-0 asset budget is $Q_A\theta_{iA} + Q_B\theta_{iB} = 0$.

Part I -- by hand. Normalize the state-B security price to $Q_B = 1$ and write $q = Q_A$. Derive individual security demands as functions of q, r_A, r_B. Show that commodity clearing under equal spot prices requires $r_A = r_B = 1$ and that the equilibrium security-price ratio is one. Calculate portfolios and state consumption, and verify both commodity and security clearing. Explain the role of strict concavity and price normalization in the uniqueness statement.

Part II -- verified scale-up. Generalize the computation to S states and a full set of Arrow securities. Build the sequential budget matrix, solve the equilibrium numerically, and verify state-by-state feasibility and zero net security supply. Ask Codex or Lean to check the linear identity mapping sequential budgets into the Arrow--Debreu present-value budget, clearly separating the algebraic equivalence from equilibrium existence.